

LAB MANUAL

COMPUTER NETWORKS

Course Code: CSCN-348



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Preface

The lab manual has been designed to familiarize students with the layered approach in data communications, switching technologies, and protocols. It would also help to have expertise in specific areas, such as the design and maintenance of various types of networks.

Tools/ Technologies

- Packet Tracer
- GNS 3

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LAB 1: Preparing Ethernet Cables

Objectives

How to wire Ethernet cables and connect them to RJ45 Jacks.

1. Tools required: -

- Cable - bulk Category 5, 5e, 6
- Wire Cutters - to cut the cable if necessary.
- Wire Stripper - to strip the cable if necessary.

a. For Patch Cables (connect switches):

- RJ45 Plugs
- RJ45 Crimper

b. For Fixed Wiring:

- RJ45 Jacks

2. Recommended:

- Wire Stripper
- Cable Tester

3. Introduction: About the Cable

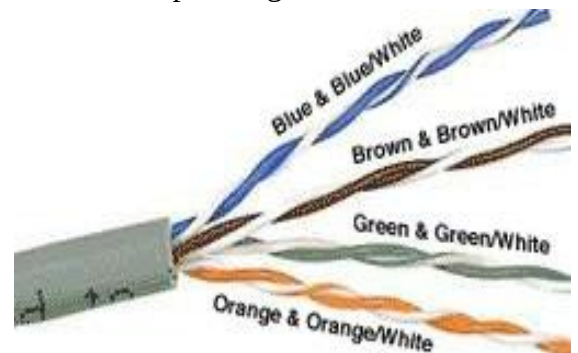
You can find bulk supplies of the cable at many computer stores or most electrical or home centers. UTP (Unshielded Twisted Pair) Category 5 cables are used for basic 10/100 Mbps functionality and CAT 5e and 6 for gigabit (1000BaseT) operation, and CAT 7 gives you a measure of future proofing.

Bulk cable comes in many types, and there are two basic categories, solid and braided cable. Braided cable tends to work better in patch applications for desktop use. It is more flexible and resilient than solid cable and easier to work with but meant for shorter lengths. Solid cable is meant for longer runs in a fixed position.

The plenum-rated cable must be used whenever the cable travels through an air circulation space, for example, above a false ceiling or below a raised floor. It may be difficult or impossible to tell from the package what type of cable it is, so peel out an end and investigate and look inside the cable. Usually, there are eight color-coded wires. These wires are twisted into four pairs of wires, and each pair has a common color theme.

One wire in the pair is a solid or primarily solid-colored wire, and the other is a primarily white wire with a colored stripe. Examples of the naming schemes used are Orange (alternatively Orange/White) for the solid-colored wire and White/Orange for the striped cable. The twists are extremely important. Twisting is there to counteract noise and interference.

It is important to wire according to a standard to get proper performance from the cable. The TIA/EIA-568-A specifies two wiring standards for an 8-position modular connector such as RJ45. The two wiring standards, T568A and T568B, vary only in the arrangement of the colored pairs.



Your choice might be determined by the need to match existing wiring, jacks, or personal preference, but you should maintain consistency. I've shown both below for straight-through cabling and just T568B for crossover cabling.

4. Introduction to RJ45 Plugs or Jacks:

An 8-pin/8-position plug or jack is commonly used to connect computers to Ethernet-based local area networks (LAN), as shown in figure 1.1. There are a couple of variations available. The primary variation students need to pay attention to is whether the connector is intended for braided or solid wire.

For braided/stranded wires, the connector has sharp pointed contacts that pierce the wire.

For solid wires, the connector has fingers that cut through the insulation and contact the wire by grasping it from both sides.

The connector is the **weak point** in an ethernet cable; choosing the wrong one will often cause grief later. If you just walk into a computer store, it's nearly impossible to tell what type of plug it is. You may be able to determine what type it is by crimping one without a cable.

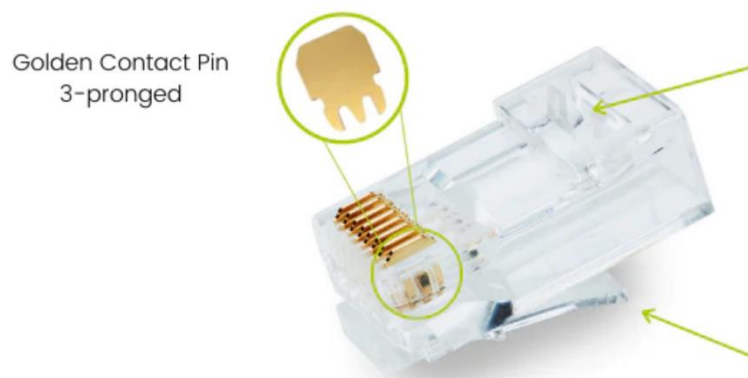


Figure 1.1 RJ45 Plug or Jack

RJ45 jacks come in a variety of styles intended for several different mounting options. The choice is one of the requirements and preferences. RJ45 jacks are designed to work only with solid cable. Most jacks come labeled with color codes for either T568A, T568B, or both.

5. Forms of Ethernet Cables - Crossed and Straight cables - When to use them

The following diagram shows the Normal use of Crossed and Straight cables (see figure 1.2).

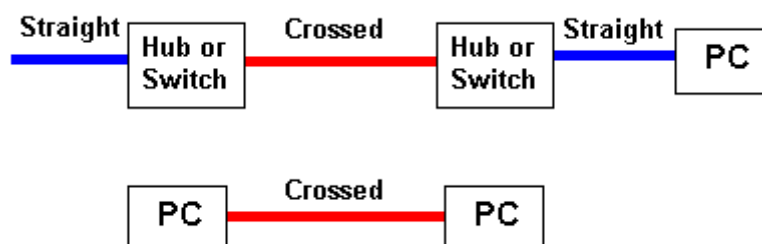


Figure 1.2 Type of ethernet cables

Notes:

- We show Straight cables as **BLUE** and Crossed as **RED**. That is our convention; the cable color can be anything you choose or, more likely, the vendor decides.
- To avoid the need for Crossed cables, many vendors provide **UPLINK** ports on Hubs or Switches - these are specially designed to allow the use of a **STRAIGHT** cable when connecting back-to-back Hubs or Switches. Read the manufacturer's documentation carefully.

6. Ethernet Cable Pin Outs and Connections

There are two basic cable pinouts. A straight-through cable is used to connect to a hub or Switch, and a crossover cable is used to operate in a peer-to-peer fashion without a hub/switch.

Generally, all fixed wiring should be run straight through. Some ethernet interfaces can cross and un-cross a cable automatically as needed, a handy feature.

NOTE: Standard, Straight-Through Wiring (both ends are the same):

Let's first learn how to find the PIN No. of the Connector. Make sure you identify and start with the correct Pin.

Where is pin #1?

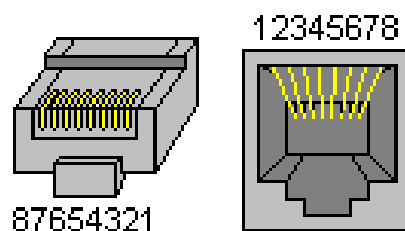


Figure 1.3 Pinout of connector

Theoretical Description**1. What Is Straight Through Cable?**

A straight-through cable is a type of twisted pair cable that is used in local area networks to connect a computer to a network hub such as a router. This type of cable is also sometimes called a patch cable and is an alternative to wireless connections where one or more computers access a router through a wireless signal. On a straight-through cable, the wired pins match. Straight-through cable uses one wiring standard: both ends use the T568A wiring standard, or both ends use the T568B wiring standard. The following figure shows a straight-through cable of which both ends are wired as the T568B standard.

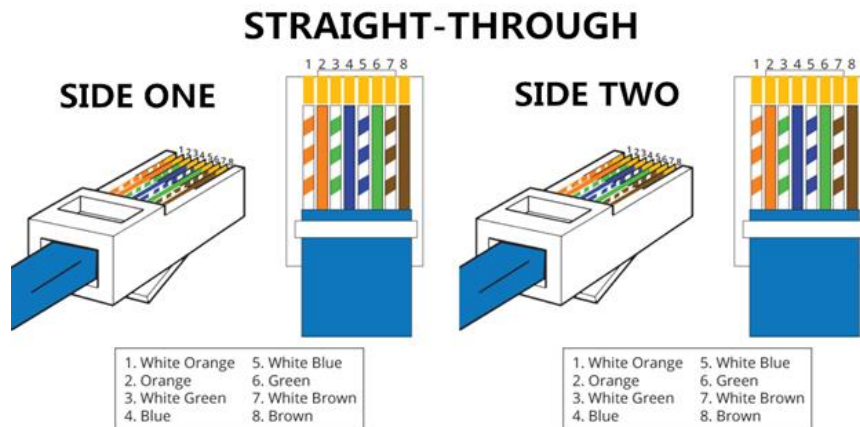


Figure 1.4 Straight-Through cable connections

2. What Is Crossover Cable?

A crossover Ethernet cable is a type of Ethernet cable used to connect computing devices directly. Unlike straight-through cable, the RJ45 crossover cable uses two different wiring standards: one end uses the T568A wiring standard, and the other end uses the T568B wiring standard. The internal wiring of Ethernet crossover cables reverses the transmit and receive signals. It is most often used to connect two devices of the same type: e.g., two computers (via network interface controller) or two switches to each other.

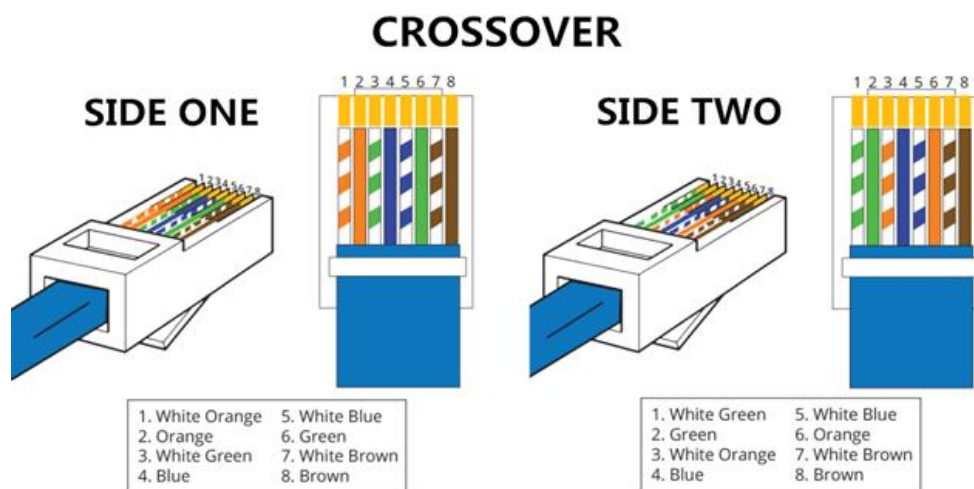


Figure 1.5 Crossover Cable

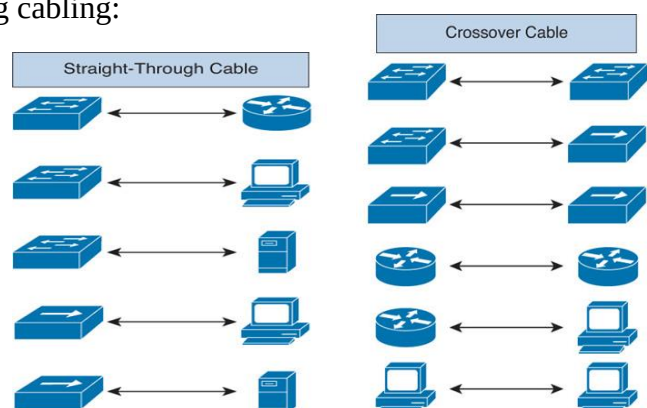
3. Straight Through vs. Crossover Cable, which to choose?

Straight through vs. crossover cable, which one should I choose? Usually, straight-through cables are primarily used for connecting, unlike devices. And crossover cables are used for connecting like devices.

Use straight-through Ethernet cable for the following cabling:

- Switch to Router
- Switch to PC or server
- Hub to PC or server

Use crossover cables for the following cabling:



- Switch to switch
- Switch to Hub
- Hub to Hub
- Router to Router
- Router Ethernet port to PC NIC
- PC to PC

4. Pin Out Tables for ethernet and fast ethernet connections

Straight-Through Cable Pin Out for T568A

RJ45 Pin #	Wire Color (T568A)	Wire Diagram (T568A)	10Base-T Signal 100Base-TX Signal	1000Base-T Signal
1	White/Green		Transmit+	BI_DA+
2	Green		Transmit-	BI_DA-
3	White/Orange		Receive+	BI_DB+
4	Blue		Unused	BI_DC+
5	White/Blue		Unused	BI_DC-
6	Orange		Receive-	BI_DB-
7	White/Brown		Unused	BI_DD+
8	Brown		Unused	BI_DD-

Straight-Through Cable Pin Out for T568B

RJ45 Pin #	Wire Color (T568B)	Wire Diagram (T568B)	10Base-T Signal 100Base-TX Signal	1000Base-T Signal
1	White/Orange		Transmit+	BI_DA+
2	Orange		Transmit-	BI_DA-
3	White/Green		Receive+	BI_DB+
4	Blue		Unused	BI_DC+
5	White/Blue		Unused	BI_DC-
6	Green		Receive-	BI_DB-
7	White/Brown		Unused	BI_DD+
8	Brown		Unused	BI_DD-

Cross Over Cable (T568B):

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Note: The crossover cable layout is suitable for 1000Base-T operation; all four pairs are crossed.

5. How to wire Ethernet Patch Cables:

- a. Strip off about 2 inches of the cable sheath.
- b. Untwist the pairs - don't untwist them beyond what you have exposed; the more untwisted cable you have, the worse the problems you can run into.
- c. Align the colored wires according to the diagrams above.
- d. Trim all the wires to the same length, about 1/2" to 3/4" left exposed from the sheath.
- e. Insert the wires into the RJ45 plug - make sure each wire is fully inserted into the front of the RJ45 plug and in the correct order. The sheath of the cable should extend into the RJ45 plug by about 1/2" and will be held in place by the crimp.
- f. Crimp the RJ45 plug with the crimper tool.
- g. Verify the wires ended up in the right order and that the wires extend to the front of the RJ45 plug and make good contact with the metal contacts in the RJ45 plug
- h. Cut the cable to length - make sure it is more than long enough for your needs.
- i. Repeat the above steps for the second RJ45 plug.

6. How to wire fixed Ethernet Cables:

- a. Run the full length (Max 100 meters) of cable in place, from endpoint to endpoint, making sure to leave excess.
- b. At one end, cut the wire to length, leaving enough length to work but not too much excess.
- c. Strip off about 2 inches of the cable sheath.
- d. Align each of the colored wires according to the layout of the jack.
- e. Use the punch-down tool to insert each wire into the jack.
- f. Repeat the above steps for the second RJ45 jack.

If a cable tester is available, use it to verify the proper connectivity of the cable. That should be it; if your cable doesn't turn out, look closely at each end and see if you can find the problem. Often a wire ends up in the wrong place, or one of the wires is making no contact or poor contact.

Also, double-check the color coding to verify it is correct. If you see a mistake or problem, cut the end off and start again. A cable tester is invaluable at identifying and highlighting these issues.

When sizing cables, remember that an end-to-end connection should not extend more than 100m (~328ft). Try to minimize the cable length; the longer the cable becomes, the more it may affect performance. This is usually noticeable as a gradual decrease in speed and increase in latency.

Lab Tasks

Task A:

Prepare the ethernet cable during the lab. Explain the steps, and attach the pictures of the equipment and steps performed with the Lab 1 report.

Task B

Perform the Lab task and attach it along with the lab report while submitting.

1. Identifying the pins on the RJ45 male connector, which are used for transmission of 100 Mbps.

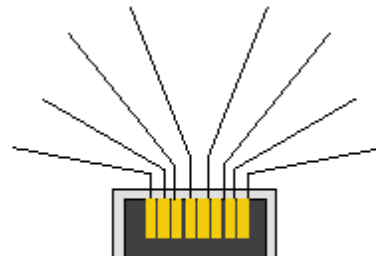
Identify The Color Pairs

PAIR NO 1 _____

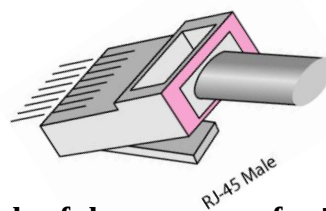
PAIR NO 2 _____

PAIR NO 3 _____

PAIR NO 4 _____



2. Correctly number the pins of the jack



2. Label the pins according to the color

3. Provide a correct color code of the connector for 568b standard

Straight Cable

1	←→	1
2	←→	2
3	←→	3
4	←→	4
5	←→	5
6	←→	6
7	←→	7
8	←→	8

Cross Cable

1	←→	1
2	←→	2
3	←→	3
4	←→	4
5	←→	5
6	←→	6
7	←→	7
8	←→	8

LAB 2: Understanding Network Commands

Objectives

- Gather information including connection, hostname, Layer 2 MAC address, and Layer 3 TCP/IP network address information.
- Compare network information to other PCs on the network.
- Learn to use the TCP/IP Packet Internet Groper (ping) command from a workstation.
- Learn to use the Trace Route (tracert) command from a workstation.
- Observe name resolution occurrences using WINS and/or DNS servers.

1. Equipment required: -

- Windows PC

Recommendation

Connectivity with internet

Theoretical Description

This lab assumes the use of any version of Windows XP and above. This is a non-destructive lab and can be done on any machine without concern about changing the system configuration. Ideally, this lab is performed in a Computer Lab or other LAN environment that connects to the internet.

The commands help us to diagnose the LAN environment and troubleshoot the network.

The following command will be used to practice during the lab:

- a. Ensuring the computer has an IP address
- b. Ping
- c. Trace Route

a. Step 1 Connect to the internet

Establish and verify connectivity to the internet. This ensures the computer has an IP address.

Step 2 Gather TCP/IP configuration information

Use the Start menu to open the Command Prompt, an MS-DOS-like window.

Press **Start > Programs > Accessories > Command Prompt**

OR

Start > Programs > Command Prompt.

OR

Press Start>Run, Then type cmd.

The following figure shows the Command screen. Type ipconfig and press the Enter key. The spelling of **ipconfig** is critical, while the case is not. It is short for IP Configuration.

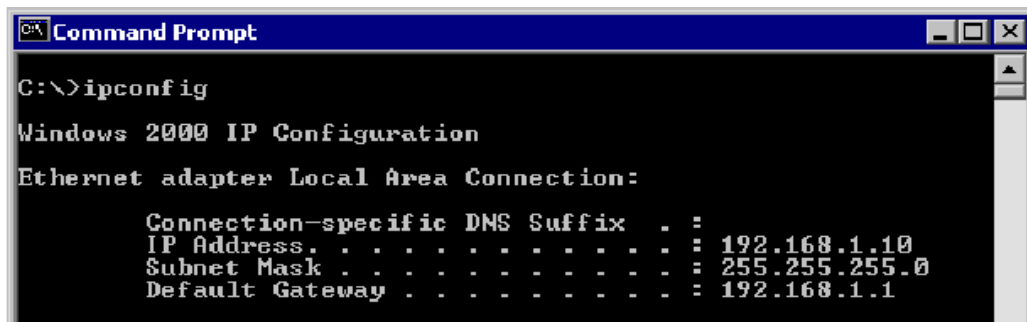


Figure 2.1 ipconfig command

This first screen shows the IP address, subnet mask, and default gateway. The IP address and the default gateway should be in the same network or subnet. Otherwise, this host would not be able to communicate outside the network. In figure 2.1, the subnet mask tells us that the first three octets must be the same to be in the same network. Further understanding shall be taught in the next lab on IP addressing.

Note: If this computer is on a LAN, the default gateway might not be seen if it is running behind a Proxy Server. Record the following information for this computer.

Step 3 Record the following TCP/IP information for this computer

IP address: _____

Subnet Mask: _____

Default Gateway: _____

Record a couple of the IP Addresses: _____

Step 5 Check additional TCP/IP configuration information

To see detailed information, type **ipconfig /all** and press **Enter**. The figure shows the detailed IP configuration screen.

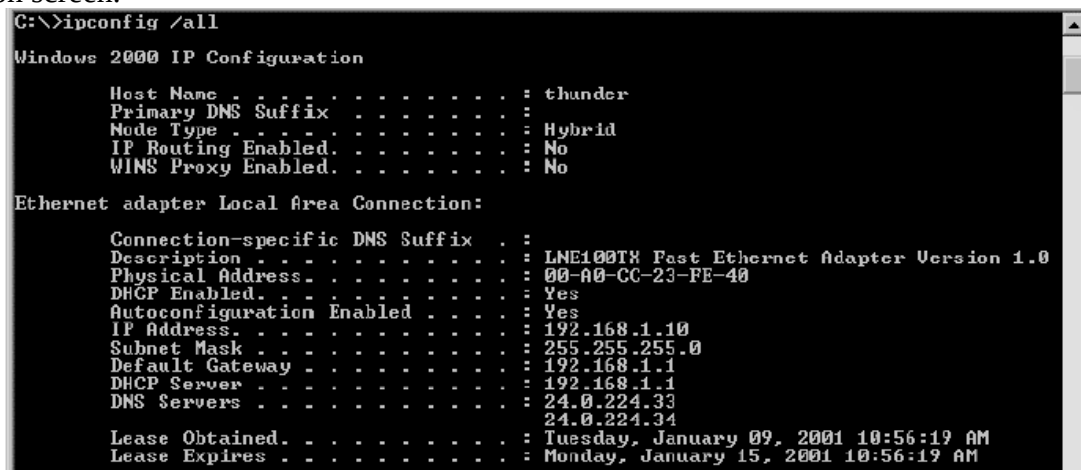


Figure 2.2 ipconfig /all

The hostname, including the computer name and NetBIOS name, should be displayed. Also, the DHCP server address, if used, and the date the IP lease starts and ends should be displayed. Look over the information. Entries for the DNS, used in name resolution servers, may also be present.

The previous figure reveals that the Router is performing both DHCP and DNS services for this network. This would likely be a small office or home office (SOHO) or small branch office implementation.

Notice the Physical Address (MAC) and the NIC model (Description).

Step 6 Close the screen

Close the screen when finished examining network settings.

Repeat the previous steps as necessary. Make sure that it is possible to return to and interpret this screen.

Reflection

Based on observations, what can be deduced about the following results taken from three computers connected to one Switch?

Computer 1

IP Address: 192.168.12.113

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.12.1

Computer 2

IP Address: 192.168.12.205

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.12.1

Computer 3

IP Address: 192.168.112.97

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.12.1

Should they be able to talk to each other? Are they all on the same network? Why or why not? If something is wrong, what is most likely the problem

b. Using "ping" and "tracert "**Objective**

- Learn to use the TCP/IP Packet Internet Groper (ping) command.
- Learn to use the Trace Route (tracert) command.
- Observe name resolution occurrences using WINS and/or DNS servers.

Background

This lab assumes the use of any version of Windows. This is a non-destructive lab and can be done on any machine without concern about changing the system configuration.

Ideally, this lab is performed in a LAN environment that connects to the internet. It can be done from a single remote connection via a modem or DSL-type connection. The student will need the IP addresses that were recorded in the previous part of the lab.

Step 1 Establish and verify connectivity to the internet

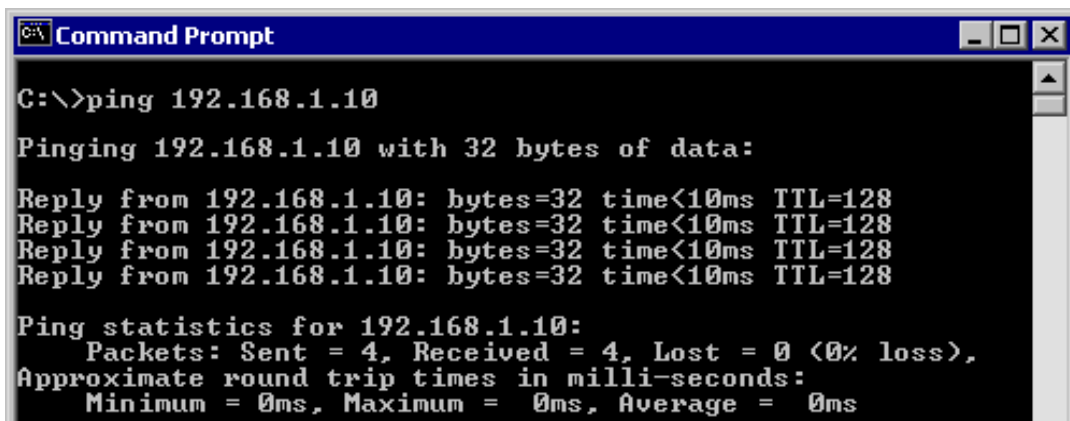
This ensures the computer has an IP address.

Step 2 Access the command prompt

As accessed in the previous part of the lab.

Step 3 ping the IP address of another computer

In the window, type **ping**, a space, and the IP address of a computer recorded in the previous lab. The following figure shows the successful results of **ping** to this IP address.



```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


Figure 2.3 Ping Command reply

ping uses the ICMP echo reply feature to test physical connectivity. Since the **ping** reports four attempts, it indicates the reliability of the connection. Look over the results and verify that the **ping** was successful. Is the **ping** successful?

Step 4 ping the IP address of the default gateway

Try to ping the IP address of the default gateway if one was listed in the last exercise. If the ping is successful, it means there is physical connectivity to the Router on the local network and probably the rest of the world.

Step 5 ping the IP address of a DHCP or DNS servers

Try to **ping** the IP address of any DHCP and/or DNS servers listed in the last exercise. If this works for either server, and they are not in the network, what does this indicate?

Was the ping successful? _____

Step 6 ping the Loopback IP address of this computer

Loopback address helps to identify if the NIC of the computer is working fine or if it is faulty.

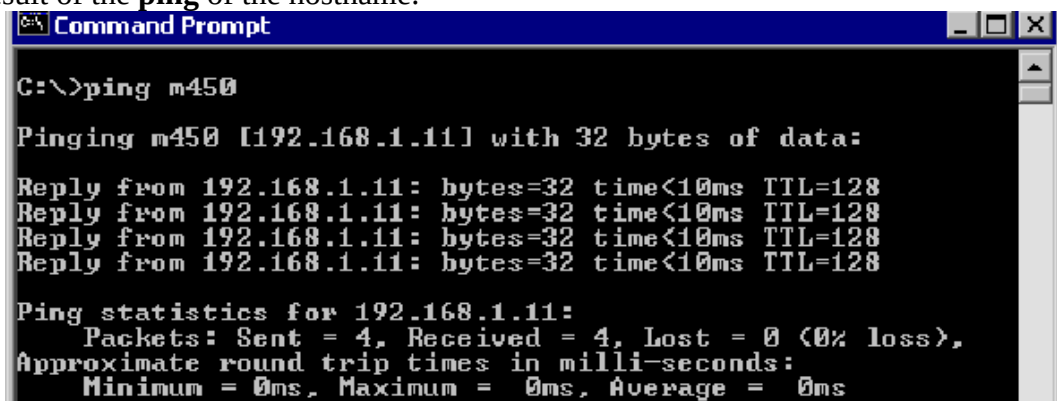
Type the following command: ping 127.0.0.1

The 127.0.0.0 network is reserved for loopback testing. If the ping is successful, then TCP/IP is properly installed and functioning on this computer.

Was the ping successful? _____

Step 7 ping the hostname of another computer

Try to **ping** the hostname of the computer that was recorded in the previous lab. The figure shows the successful result of the **ping** of the hostname.



```

C:\>ping m450

Pinging m450 [192.168.1.11] with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

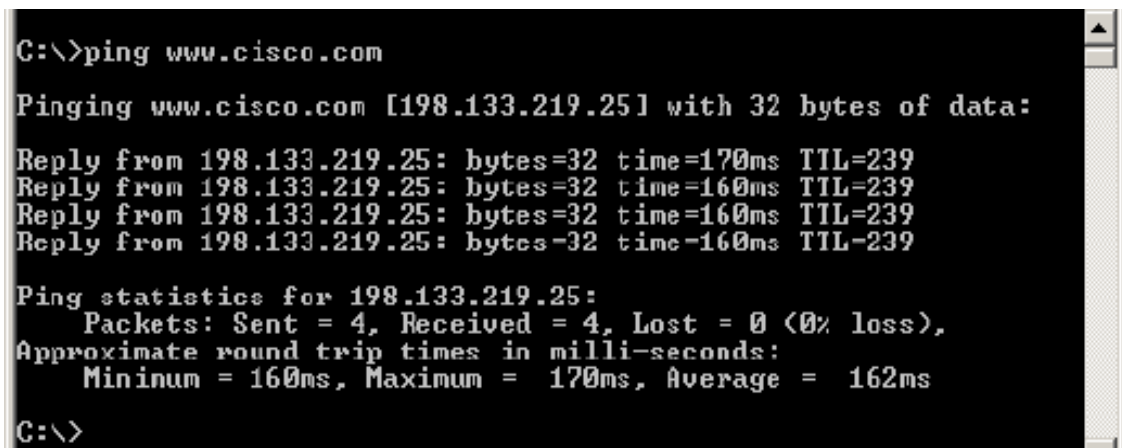
Figure 2.4 Pinging hostname

Look over the results. Notice that the first line of output shows the hostname, m450, in the example, followed by the IP address. This means the computer was able to resolve the hostname to an IP address. Without name resolution, the **ping** would have failed because TCP/IP only understands valid IP addresses, not names.

If the **ping** was successful, it means that connectivity and discovery of IP addresses can be made with only a hostname. This is how many early networks communicated. If successful, then **ping** a hostname also shows that there is probably a WINS server working on the network.

Step 8 ping the Cisco website

Type the following command: **ping www.cisco.com**



```

C:\>ping www.cisco.com

Pinging www.cisco.com [198.133.219.25] with 32 bytes of data:

Reply from 198.133.219.25: bytes=32 time=170ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239

Ping statistics for 198.133.219.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 160ms, Maximum = 170ms, Average = 162ms

C:\>

```

Figure 2.5 Ping Cisco.com

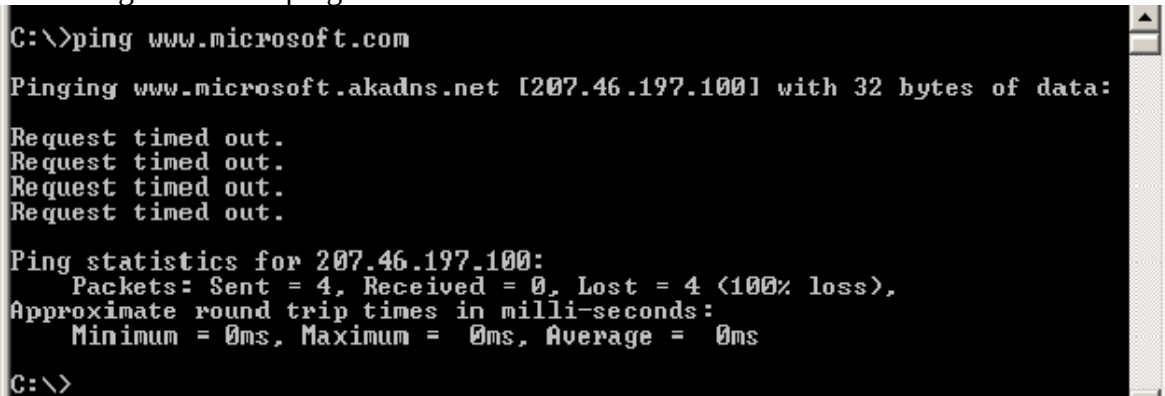
The first output line shows the Fully Qualified Domain Name (FQDN) followed by the IP address. A Domain Name Service (DNS) server somewhere in the network was able to resolve the name to an IP address. DNS servers resolve domain names, not hostnames, to IP addresses.

Without this name resolution, the ping would have failed because TCP/IP only understands valid IP addresses. It would not be possible to use the web browser without this name resolution.

With DNS, connectivity to computers on the internet can be verified using a familiar web address, or domain name, without having to know the actual IP address. If the nearest DNS server does not know the IP address, the server asks for a DNS server higher in the Internet structure.

Step 9 ping the Microsoft website

Type the following command: ping www.microsoft.com



```

C:\>ping www.microsoft.com

Pinging www.microsoft.akadns.net [207.46.197.100] with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 207.46.197.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 2.6 ping Microsoft.com

Notice that the DNS server was able to resolve the name to an IP address, but there is no response. Some Microsoft routers are configured to ignore ping requests. This is a frequently implemented security measure.

Ping some other domain names and record the results. For example, **ping www.msn.de**

c. Trace the route to the Cisco website

Type **tracert www.cisco.com** and press **Enter**.

tracert is TCP/IP abbreviation for traceroute. The preceding figure shows the successful result when running a **tracert** from Bavaria in Germany. The first output line shows the FQDN followed by the IP address. Therefore, a DNS server was able to resolve the name to an IP address. Then there are listings of all routers the **tracert** requests had to pass through to get to the destination.

```

C:\>tracert www.cisco.com

Tracing route to www.cisco.com [198.133.219.251]
over a maximum of 30 hops:
  0  <10 ms    <10 ms    <10 ms    10-37-00-1.internal.alp.dillingen.de [10.37.0.1]
  1  <10 ms    <10 ms    <10 ms    194.95.207.11
  2  <10 ms    <10 ms    <10 ms    ar-augsburg2.g-win.dfn.de [188.1.37.145]
  3  <10 ms    <10 ms    <10 ms    ar-augsburg1.g-win.dfn.de [188.1.74.193]
  4  <10 ms    <10 ms    <10 ms    cr-muenchen1.g-win.dfn.de [188.1.74.33]
  5  <10 ms    <10 ms    <10 ms    cr-frankfurt1.g-win.dfn.de [188.1.18.81]
  6  <10 ms    <10 ms    <10 ms    so-6-0-0.ar2.FRA2.gblx.net [208.48.23.141]
  7  <10 ms    <10 ms    <10 ms    pos3-0-622M.cr1.FRA2.gblx.net [62.16.32.73]
  8  <10 ms    <10 ms    <10 ms    soc0-0-0-2488M.cr2.LON3.gblx.net [195.8.96.174]
  9  <10 ms    <10 ms    <10 ms    pos1-0-622M.br1.LON3.gblx.net [195.8.96.189]
 10  <10 ms    <10 ms    <10 ms    sl-bb21-lon-5-0.sprintlink.net [213.206.131.251]
 11  <10 ms    <10 ms    <10 ms    sl-bb20-msq-10-0.sprintlink.net [144.232.19.69]
 12  <10 ms    <10 ms    <10 ms    sl-bb20-rlg-15-1.sprintlink.net [144.232.19.94]
 13  <10 ms    <10 ms    <10 ms    sl-bb22-sj-5-1.sprintlink.net [144.232.9.125]
 14  <10 ms    <10 ms    <10 ms    sl-bb25-sj-12-0.sprintlink.net [144.232.3.210]
 15  <10 ms    <10 ms    <10 ms    sl-gw11-sj-10-0.sprintlink.net [144.232.3.134]
 16  <10 ms    <10 ms    <10 ms    sl-ciscopen2-11-0-0.sprintlink.net [144.228.44.141]
 17  <10 ms    <10 ms    <10 ms    sjck-dirty-gw1.cisco.com [120.107.239.5]
 18  <10 ms    <10 ms    <10 ms    sjck-sdf-ciad-gw1.cisco.com [128.107.239.106]
 19  <10 ms    <10 ms    <10 ms    www.cisco.com [198.133.219.251]
 20  <10 ms    <10 ms    <10 ms    www.cisco.com [198.133.219.251]

Trace complete.

```

tracert uses the same echo requests and replies as the ping command but in a slightly different way. Observe that tracert contacted each Router three times. Compare the results to determine the consistency of the route. Notice in the above example that there were relatively long delays after routers 11 and 13, possibly due to congestion. The main thing is that there seems to be a relatively consistent connection.

Lab Tasks

1. Ping 127.0.0.1
2. Ping the default gateway
3. Ping your hostname
4. Ping the DNS server(s)

Do you get a positive reachability result? _____

5. How to obtain the following network configuration on your computer.

Host Name:

MAC Address:

IP Address:

Subnet Mask:

Default Gateway:

DHCP Servers:

Open a web browser and visit the following website: <http://whatismyipaddress.com/>
Is the reported IP address consistent with that obtained by ipconfig? If not, why?

LAB 3: IP Addressing and Subnetting

Objectives

- Study the difference between IPv4 and IPv6.
- Learn the concept of subnetting.
- Study fixed length and variable length subnetting in IPV4.

Theoretical Description

IP Addressing

The Internet Protocol (IP) is the method or protocol by which data is sent from one computer to another on the LAN or WAN. Each computer (known as a host) on the internet has at least one IP address that uniquely identifies it from all other computers on the internet. Each Network Interface Card (NIC or Network card) present in a PC is assigned one Network address called an IP address. This IP address is assigned by the administrator of the network. There is a burned-in address on the NIC called a Physical Address [or MAC]. The MAC address of a network card indicates the vendor of that card and a unique serial number.

IPv4	IPv6
IPv4 addresses are 32-bit in length.	IPv6 addresses are 128-bit in length.
IPv4 addresses are binary numbers represented in decimals.	IPv6 addresses are binary numbers represented in hexadecimal.
IPsec support is only optional.	Inbuilt IPsec support.
Fragmentation is done by the sender and forwarding routers.	Fragmentation is done only by the sender.
Broadcast Multicast Uni-cast	Uni-cast Multicast Any-cast
Manual configuration (Static) of IPv4 addresses or DHCP (Dynamic configuration) is required to configure IPv4 addresses.	Auto-configuration of addresses is available.

Roles of IPv4 Addressing

Format of the IP address

IPv4 is made up of four parts, in the pattern of w.x.y.z. Each part has eight binary bits, and the values in decimal can range from 0 to 255. For example 31.223.182.162

IP address classes

IP addresses are divided into different classes. These classes determine the maximum number of hosts per network ID. Only three classes are used for network connectivity.

The following table lists all the address classes.

IP address class	Value of the leftmost byte (part w)	1 st Octet
Class A	1-126 [0 and 127 are not allowed]	0
Class B	128-191	10
Class C	192-223	110
Class D	224-239	1110
Class E	240-255	1111

Grouping of IP addresses into different classes: A, B, C, D, and E.

Class A: first bit in part w is 0, and others can be anything.

IP Addressing

0000 0001: 1

0111 1110: 126

The first 8 bits are used for the network part, and the remaining for the host part.

Class B: First two bits in w are 10, and the others can be anything

1000 0000: 128

1011 1111: 191

The first 16 bits are used for the network part, and the remaining for the host part

Class C: first three bits in w are 110, and others can be anything

1100 0000: 128

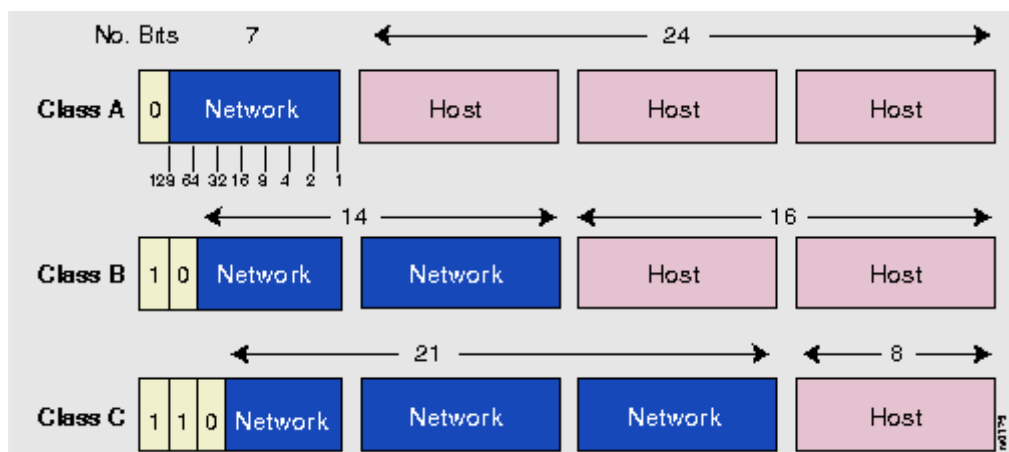
1101 1111: 223

The first 24 bits are used for the network part, and the last 8 bits for the host part.

Class D: first four bits in w are 1110, and others can be anything; used for multicast.

1110 0000: 224

1111 1111: 239



Default Subnet mask

It is used to identify the network part from the host part. Put binary one for the parts that represent the network part and zero for the part that represents the host part.

Address Class	Bits for Subnet Mask			Subnet Mask
Class A	11111111 00000000	00000000	00000000	255.0.0.0
Class B	11111111 00000000	11111111	00000000	255.255.0.0
Class C	11111111 00000000	11111111	11111111	255.255.255.0

Private IP Addresses

Class	Private Networks	Subnet Mask	Address Range
A	10.0.0.0	255.0.0.0	10.0.0.0 - 10.255.255.255
B	172.16.0.0 - 172.31.0.0	255.240.0.0	172.16.0.0 - 172.31.255.255
C	192.168.0.0	255.255.0.0	192.168.0.0 - 192.168.255.255

Invalid IP address:

- If the network part is all 0s, the address belongs to the class, but this is an invalid IP address because, for an IP address, all the network or host parts should not be all 1s or all 0s.
- If the network part is all 1s, this address belongs to class E. But due to the presence of all 1s, it is not valid. This represents broadcast to all networks.
- If the host part is all 0s, this represents the network address. This is not a valid IP address.
- If the host part is all 1s, this represents the broadcast address. This is not a valid IP address.
- 127.0.0.0 network address is used for loopback testing. This will help you to check the network card of your own PC [localhost].
- The validity of the IP address is also based on the subnet mask used provided.

Unicast: one device calls another.

Multicast: one device calls a group of the network.

Broadcast: one device calls the entire network.

Anycast: Mostly near

Subnetting

Subnetting is the practice of dividing a network into two or more networks.

Subnet mask

The subnet mask is a 32-bit value that allows the receptionist of the IP packet to distinguish the network ID portion in the IP address from the host ID portion.

Class A, B, and C networks have default masks, also known as natural masks, as shown here:

Class A: 255.0.0.0

Class B: 255.255.0.0

Class C: 255.255.255.0

Subnetting

Subnetting is the process of designating some high-order bits from the host part and grouping them with the network mask to form the subnet mask. This divides a network into smaller subnets. The following diagram modifies the example by moving 2 bits from the host part to the subnet mask to form four smaller subnets one-quarter the previous size:

	Binary form	Dot-decimal notation
IP address	11000000.10101000.00000101.10000010	192.168.5.130
Subnet mask	11111111.11111111.11111111.11000000	255.255.255.192
Network prefix	11000000.10101000.00000101.10000000	192.168.5.128
Host part	00000000.00000000.00000000.00000010	0.0.0.2

Special addresses and subnets

Internet Protocol version 4 uses specially designated address formats to facilitate recognition of special address functionality. The first and the last subnets obtained by subnetting have traditionally had a special designation and, early on, special usage implications. In addition, IPv4 uses all one's host addresses, i.e., the last address within a network, for broadcast transmission to all hosts on the link.

Subnet zero and the all-ones subnet

The first subnet obtained from subnetting has all bits in the subnet bit group set to zero. It is therefore called subnet zero. The last subnet obtained from subnetting has all bits in the subnet bit group set to one. It is therefore called the all-ones subnet.

The IETF originally discouraged the production use of these two subnets due to possible confusion about having a network and subnet with the same address. The practice of avoiding subnet zero and the all-ones subnet was declared obsolete in 1995 by RFC 1878, an informational but now historical document.

Subnet and host counts

The number of subnetworks available and the number of possible hosts in a network may be readily calculated. In the example (above), two bits were borrowed to create subnetworks, thus creating 4 (2²) possible subnets.

Network	Network (binary)	Broadcast address
192.168.5.0/26	11000000.10101000.00000101.00000000	192.168.5.63
192.168.5.64/26	11000000.10101000.00000101.01000000	192.168.5.127
192.168.5.128/26	11000000.10101000.00000101.10000000	192.168.5.191
192.168.5.192/26	11000000.10101000.00000101.11000000	192.168.5.255

The RFC 950 specification recommended reserving the subnet values consisting of all zeros (see above) and all ones (broadcast), reducing the number of available subnets by two. However, due to the inefficiencies introduced by this convention, it was abandoned for use on the public internet and is only relevant when dealing with legacy equipment that does not implement CIDR. The only reason not to use the all-zeroes subnet is that it is ambiguous when the prefix length is not available. RFC 950 itself did not make the use of the zero-subnet illegal; it was, however, considered best practice by engineers.

CIDR-compliant routing protocols transmit both length and suffix. RFC 1878 provides a subnetting table with examples.

The remaining bits after the subnet bits are used for addressing hosts within the subnet. In the above example, the subnet mask consists of 26 bits, leaving 6 bits for the host identifier. This allows for 62 host combinations (2⁶-2).

The all-zeros value and all-ones values are reserved for the network address and broadcast address, respectively. In systems that can handle CIDR, a count of two is therefore subtracted from the host

availability rather than the subnet availability, making all 2^n subnets available and removing a need to subtract two subnets.

For example, under CIDR /28, all 16 subnets are usable. Each broadcast, i.e., .15 .31 - .255, comes off the client count, not the network, thus making the last subnet also usable.

Only legacy technology not capable of using CIDR default in accordance with the RFC 1878 standard required the subtraction of subnets, one at the beginning of the range and one at the end of the range. Cisco added to this confusion by the use of subtracting from the subnetting formula in its publications for so-called usable subnets up until 2007. Cisco routers, by default, did not allow an IP address belonging to subnet zero to be configured on an interface.

In general, the number of available hosts on a subnet is $2^h - 2$, where h is the number of bits used for the host portion of the address. The number of available subnets is 2^n , where n is the number of bits used for the network portion of the address. This is the RFC 1878 standard used by the IETF, the IEEE, and COMPTIA. RFC 3021 specifies an exception to this rule for 31-bit subnet masks, which means the host identifier is only one bit long for two permissible addresses. In such networks, usually point-to-point links, only two hosts (the endpoints) may be connected, and a specification of network and broadcast addresses is not necessary. A /24 network may be divided into the following subnets by increasing the subnet mask successively by one bit. This affects the total number of hosts that can be addressed in the /24 network (last column).

Prefix size	Network mask	Available subnets	Usable hosts per subnet	Total usable hosts
/24	255.255.255.0	1	254	254
/25	255.255.255.128	2	126	252
/26	255.255.255.192	4	62	248
/27	255.255.255.224	8	30	240
/28	255.255.255.240	16	14	224
/29	255.255.255.248	32	6	192
/30	255.255.255.252	64	2	128
/31	255.255.255.254	128	2 *	256

*Only applicable for point-to-point links

Method of subnetting:

There are two methods of subnetting.

- FLSM (Fixed Length Subnet Mask)

In this method, the network is divided into equal lengths.

For example, if the required hosts are 50

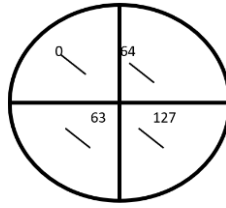
We divide the network into

0-63

64-127

128-191

192-256



- VLSM (Variable Length Subnet Mask)

In this method, the network is divided into different lengths.

Example of subnetting in class C:

Class C	Network	Network	Network	Host
Subnet Mask	255	255	255	0

For example, the required host is 60 from the below address

IP Address 192.168.1.0/24 255.255.255.0

1. After a search in the given table $64 < 60 > 32$. This means $2^6 = 64$ is the required host, so $H=6$
2. **Converted network bits** = Total host bits – Required host bits = $8 - 6 = 2$. $N=2$.

As we are using class C then we have 24 network bits.

$$\begin{aligned}
 2^0 &= 1 \\
 2^1 &= 2 \\
 2^2 &= 4 \\
 2^3 &= 8 \\
 2^4 &= 16 \\
 2^5 &= 32 \\
 2^6 &= 64 \\
 2^7 &= 128 \\
 2^8 &= 256
 \end{aligned}$$

3. **Total Network bits** = Total + Converted network bits = $24 + 2 = 26$.
4. Now we can find subnet mask by adding 1,s in to the two least significant bits **255.255.255. x x x x x x x x**

128 64 32 16 8 4 2 1
1 1 0 0 0 0 0 0

We add first two enable bits.

128 + 64 = 192

The Network address and subnet mask is 192.168.1.0 /26 255.255.255.192.

1. Block size of the network:
 2. 0 to 63 192.168.1.0 (Network ID for 1st Block)-192.168.1.63(Broadcast ID for 1st Block)
 3. 64 to 127 192.168.1.64 - 192.168.1.127
 4. 128 to 191 192.168.1.128 - 192.168.1.191
- 192 to 255 192.168.1.192 - 192.168.1.255

Lab Tasks

1. What address Class does 172.16.33.1/24 belong to?
2. When calculating usable hosts per subnet, the following formula is used $2^{\text{bits}} - 2$. For what reason is two subtracted?

LAB 4: Configuring basic Switch

Objectives

- Perform basic switch configuration in Cisco packet tracer to enable Switch for the basic functionality.
- Complete walk-through of 100 MB full duplex configuration in Cisco Switch step by step.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

A good thing about the Cisco IOS is that it works quite similarly on both the routers as well as switches. However, there is a difference in the commands that we use for each one of them. For example, some specific commands are meant only for the switches while others for routers as per the device's functionality. In this lab, we will assign an IP address to the Vlan1 interface and we will configure the default gateway on the Switch. We will also perform some basic configurations which will prepare the Router for the functional network.

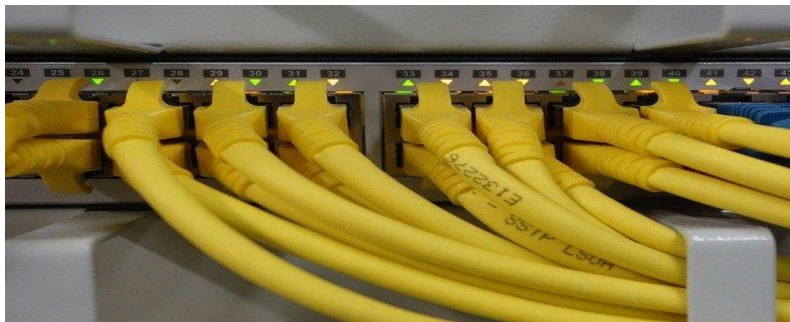


Figure 4. 1 Ethernet cables connected to the Switch

Introduction to basic Switch configuration:

All switches come with default VLAN1. To assign an IP address to the VLAN1 interface, we have to enter into interface configuration mode. While configuring the Switch for basic functionality, we should also assign a default gateway to the Switch.

It is important to remember that the VLAN1 interface and default gateway IP address must be on the same network. This enables the Switch to redirect traffic to other networks.

Saving the configurations on Switch

Vlan1 interface allows to connect with the Switch remotely via telnet, we can use the telnet client to remotely manage and configure the Switch. If you are unable to save the changes made to the Switch, all the hard work will do down the drain. Therefore, if the changes made to the Switch is worth-a-while and nothing unexpected has occurred, the user may save configuration on Cisco devices by typing the command 'write'. It copies the running-config to the startup-config alternatively we can also use the command 'copy running config startup config' to save the running configuration of the device.

No IP Domain Lookup

While configuring Cisco devices, it's common to make typing mistakes. If we mistype a command then the Cisco device will try to resolve the name to the IP address. This makes Cisco devices freeze for about 60

seconds as it resolves IP addresses in the background. One minute is the maximum time taken by the device. Now, we have to wait until the device allows us to type something again. This can be very annoying. It happens because the domain name lookup is enabled by default in Cisco devices. Hence, to solve this problem we can disable domain lookup by entering the command 'No IP Domain Lookup' in switch command line.

Logging synchronous

Now, let us talk about the status messages received on Cisco devices. When we receive status messages on Cisco devices the message kind of mixes with the text that we type on the CLI. This can also be very confusing sometimes. So, to solve this problem we can type the command 'logging synchronous'. This command synchronizes the status messages. Hence, the message does not interfere with our typing.

Speed and Duplex:

100Mb Full Duplex configuration.

Description:

Simplex – In this mode of communication, data is sent only in one direction, and the device at the receiving end will only receive the data. The role of the sender and receiver cannot be changed in this mode of connection.

Example of Simplex – TV is an example of simplex communication, it only receives data and no data is sent back in the opposite direction.

Half-duplex – On this mode switch can only receive or send at the same time hence half duplex is a slow connection and prevent the full-duplex Switch to work with full potential. On the other hand, hubs do not have full-duplex functionality so they can only receive or send data at once.

Example of half-duplex – Radio is an example of a half-duplex in which data is sent by one device and received by the other.

Full-duplex – full-duplex functionality allows a switch to transfer and receive data at the same time that makes the bandwidth double therefore, we must check the devices for duplex mismatch because duplex mismatch causes a collision and when a collision takes place then the data is lost so the device has to send the data again which further reduces the speed of the network.

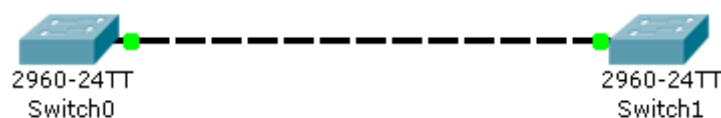
Example of the full-duplex – Telephone is an example of full-duplex communication as data can be sent and received at the same time.

Most of the time Cisco auto-negotiation for duplex selection works perfectly, however in some instances, a Cisco device is unable to detect the duplex mode and it may result in a duplex mismatch. This could result in collisions in the network. Therefore, to fully utilize the bandwidth and the capacity of the Switch, it is a good practice to hard-code the speed and duplex on the Cisco devices.

In this lab, we will configure the speed to 100 Mbps and the duplex mode to full. The duplex mode should be configured on both switches. Now, to configure the duplex and speed, we will have to go into the specific interface on which we want to hard-code the duplex setting.

After configuring both switches with full-duplex, we can solve the issue that occasionally occurs due to auto-negotiate settings in the switches and if there is a 100MB connection between two switches then full-duplex allow switches to work with double speed.

Please note that full-duplex configuration must be configured on each interface on which we want to hardcode the full-duplex mode.



Lab Tasks

Task 1

```
Switch(config)#interface fastethernet 0/1  
Switch(config)#duplex full
```

Task 2

```
Switch(config)#interface fastethernet 0/1  
Switch(config)#speed 100
```

Task 3

Apply the above on all ports of Switch, How Use range command

Task 4

```
Switch(config)#interface vlan 1  
Switch(config-if)#ip address 192.168.4.2 255.255.255.0
```

Task 5

```
Switch(config)#ip default-gateway 192.168.4.1
```

Task 6

```
Switch#write
```

Task 7

```
Router(config)#no ip domain lookup
```

Task 8

```
Switch(config)#line con 0  
Switch(config-line)#logging synchronous
```

LAB 5: Securing the Switch

Objectives

- Configuring MOTD banner and Telnet in Packet Tracer.
- Learn about telnet configuration in Cisco packet tracer
- We will encrypt various passwords configured on the Cisco device using one command.
- we will be configuring switch port security in packet tracer to secure the Switch from unauthorized access. We will allow only one MAC address to connect on switch-port.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

MOTD banner and Telnet

This lab exercise entails the basic setup of the Switch or a router. In this lab, we will configure the IP address on a fast Ethernet port of the Router. We will name the interface the 'Branch Office Interface.' Naming the interface is an excellent and very essential practice because it will let us know, to which location the port is connected. As a Network Admin, when we log into a router, the description of the interfaces is very helpful. It makes management of the interfaces a wee bit easier especially in the big networks where there are many different devices. The user has to go into the interface configuration mode to set up the IP Address and description.

You will also learn to configure MOTD [Message of the Day in this lab. When the user is logged into the device then a message appears on the screen. That message is known as the Message of the Day (MOTD). Usually, the MOTD banner is set up as a warning against unauthorized access.

Some people show ASCII art and huge messages like terms and conditions as a message of the day however best practice is to use short and precise MOTD banners as big messages put extra load on the device which results in unnecessary processor cycles.

It's all up to the admin to choose whatever message of the day should show up when we log in to the Cisco device.

Telnet connection and Testing

Configuring the Telnet Connection on the Router allows the user to manage the Router remotely with the help of a Telnet Protocol. Setting up a Telnet Connection also requires a Telnet Password to be configured. We will configure that password in this lab. Also, we will have to get into the Telnet lines to configure the telnet. We can do this by typing the line VTY command.

Configuration of the telnet helps a lot when we have to manage the device from the remote location or we are not available onsite. Configuration of the devices remotely has become common practice now a day and especially due to the Coronavirus outbreak, most of the IT individuals are working from home using the internet.

Using the internet is not to connect to an office network is not safe and secure so to overcome this issue we should use a VPN connection or we should use the SSH instead of telnet as it encrypts the data which prevents the network from attacks of the hackers.



Telnet configuration in Cisco packet tracer.

Description:

Telnet -Protocol developed in 1969, allow us to connect to the remote device using the command line interface, telnet protocol is part of the TCP/IP suite.

To use the telnet, the device should have a telnet client installed and the remote device should be configured to accept telnet connection so most of the devices like servers, routers, switches, firewalls, etc are configured to allow telnet connection.

We can use any telnet client to establish the connection; one of the famous free telnet clients is putty. Putty is used by many organizations as it is open source and it also works for the SSH connection.

In this packet tracer lab, we will set up a router for telnet access. And then use the command prompt on the computer to test our telnet connection.

To telnet the Router from our PC we will have to assign an IP address to our PC. After that, we will assign the IP address to the router interface which is connected with that PC.

For establishing a successful telnet connection, our PC and Router should be on the same network as the PC and Router are connected directly.

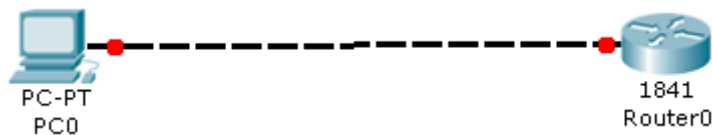
After assigning the IP addresses to both our devices, we have to enable the Router for a telnet connection by configuring the VTY lines.

We will configure telnet lines 0-15 on the Router, which will allow 16 simultaneous telnet connections to the Router, this feature is helpful if more than one person wants to log in to the device at the same time, this type of practice is common if we have many administrators to look after the device.

Once we have configured both devices properly, we can try pinging the Router from our PC to check the connectivity. If the Router is responding to the ping command then we can establish a telnet connection from our PC's command prompt.

Once the telnet connection is initiated successfully, we will be prompted with the password to access the Router via a telnet connection.

An important thing, we have to configure the privileged mode password. That's because it is not possible to access the privileged mode through a telnet connection if the password for our privileged mode is not configured. if we will try to access privileged mode without the configuration of the password then we will see an error message saying 'no password set'.



Cisco Password Encryption.

Configuring passwords and securing up devices is very important and we must configure the password on different layers to prevent unauthorized access to the device.

Due to the increased number of cyber-attacks, networks now a day are more prone to online attacks rather than physical damage to the devices and hackers continuously try to find out the loophole in the network to compromise the security. Every year organizations lose thousands of dollars due to cyber-attacks and sometimes important data is lost permanently. Most of these attacks can be prevented if the security is properly implemented at every layer.

There are various types of devices available in the market to stop these attacks like advance firewalls, intrusion prevention systems, antivirus, etc however we should not ignore the basic security parameters like

setting up a good password on the device. Organizations spend a large amount of money on expensive hardware to secure the networks however that can be of no use if security is not implemented at every layer. In this lab, we will encrypt all the passwords on the device with one command.

When we configure passwords like Privileged Mode password, VTY line password, Console line password, etc. on our Cisco devices, these passwords are visible as a clear text in the running-config. This is a security vulnerability since anyone can view them just by looking at the running-config or start-up config. Even when you are checking the Cisco device configuration, someone could be watching over your shoulder. To fix this problem, we could use a command known as the 'service password-encryption'. This command encrypts all passwords and converts them into alphanumeric numbers. Hence, we can use this command to store and display the passwords in encrypted form for added security.

Service password-encryption command encrypts the password so just by looking at the password, no will be able to figure out or steal it however it does not provide complete protection as someone can easily copy that encrypted password from the running or startup-config and can find the used password by decrypting that, there are some sites available online that easily decrypt the password with just one click so we must use enable secret that use MD5 hash to encrypt the password which is a very powerful encryption and it is not easy to reverse engineer the hash. Although password encryption should be enabled by default but it is not so we have to manually encrypt the passwords.

Cisco Port Security

Configuring switch port security in packet tracer.

Cisco has implemented some security measures to safeguard its devices from unauthorized access. Therefore, to be able to secure the physical ports on a Cisco Switch, we have to enable Port Security.

There are a few options available to the user while configuring the Port Security. One such option is to limit the number of MAC Addresses that the port can connect with. This security feature does not provide complete security however it makes it extremely difficult for the attacker to break into the network. The only way a hacker can beat this security parameter is by finding out the attached mac-address and giving the same mac-address to the device which attackers want to connect to the particular switch port.

Many times, some employees in the organization connect their devices to the switch port available in their cubicle just for the fun or sometimes for experimental purposes, and if the security is not implemented then that device can jeopardize the whole network and hackers can also this vulnerability to hack into the network.

In this lab, we will allow only one MAC Address to be able to access the interface fast Ethernet 0/2. If some other device is attached to the interface the port will go to a shutdown state. Also, when the port will be shut down then the light on the Switch will turn amber on that switch port so the admin can find out the issue just by looking at the physical indicator on the Switch.

To enable Port Security on a Cisco switch, we have to change the 'Switchport' mode from dynamic to access. By default, the Switchport mode is set to dynamic. Once the Switchport mode is changed, we can then enable the Switchport security on that port.

There are couple of ways in which we can restrict devices with random mac-address to connect with the switch port.

One way is to hardcode the mac address on the switch interface with the command "switchport port-security mac-address [mac-address number]"

Another command that we can use is "switchport port-security mac-address sticky", this command will stick the already attached mac-address to the interface and other mac-address will be rejected so if devices with other mac-address will try to connect with the interface then the interface will simply reject the connection and we can also configure how the interface should react when the security violation takes place.

Lab Tasks

Task 1

Click on pc2 > click desktop tab > click on terminal > click ok
(you should be able to access Router now)

Task 2

```
Router>
Router>enable
Router#configure terminal
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
```

Task 3

```
Router(config)#interface fastEthernet 0/0
Router(config-if)#description BRANCH OFFICE INTERFACE
```

Task 4

```
Router(config)#banner motd / --- UNAUTHORIZED ACCESS IS NOT ALLOWED --- /
```

Task 5

```
Router(config)#line vty 0 15
Router(config-line)#password cisco
Router(config-line)#login
```

Task 6

Click on pc0 > click on desktop tab > click on ip configuration > set ip address

Task 7

```
Router(config)#interface fastethernet 0/0 Router(config-if)#ip address 192.168.1.1 255.255.255.0
```

Task 8

```
R1(config)#enable password cisco Task Router(config)#line vty 0 15 Router(config-line)#password cisco
Router(config-line)#login
```

Task 9

Click on pc0 > click on command prompt > type telnet 192.168.1.1 > type privileged mode password

Task 10

```
Router(config)#enable secret cisco
```

Task 11

```
Router(config)#service password encryption
```

Task 12

```
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
```

Task 13

```
Switch(config-if)#switchport port-security violation shutdown
```

LAB 6: Virtual LAN (VLANs)

Objectives

- Create and configure VLANs in Cisco packet tracer.
- Study the Router on a Stick Configuration.
- Learn to Configure native VLAN in Cisco packet tracer and Fix native VLAN mismatch.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

We will create VLAN 10 and VLAN 20 in this lab. It is always a good practice to give names to the VLANs as this makes it easier for the admins to manage the configured VLANs. The best way to give a name is according to their role in the network for e.g. if there is a VLAN that handles traffic for the voice of IP then we can give the name 'voice' to the VLAN and another way of assigning a name is according to the departments in the organization like Sales, marketing, etc.

In a case of a network outage or any issue with the VLANs, admins can easily identify the VLANs with names, which makes their work easier

We can use the following command to create VLAN 10 and 20 and give it a name.

```
Switch(config)#vlan 10
```

```
Switch(config-vlan)#name Voice
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name Sales
```

Once we are done creating the VLANs, we can check the created VLANs with the command 'show VLAN'

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Voice	active	
20	Sales	active	

Assigning ports to VLANs:

You can see in the image above that although we have created the VLANs however to use those VLANs, we have to assign ports to VLAN. By default, all ports are assigned to VLAN 1

We will assign 1 port to VLAN 10 and 5 ports to VLAN 20

If we want to assign multiple ports to a single VLAN then we can use the interface range command and include the range of ports that we want to add to any VLAN

```
Switch(config)#interface fastEthernet 0/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 10
```

```
Switch(config)#interface range fastEthernet 0/2 – 6
```

Switch(config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 20

Now, we can see in the image below that assigned ports are appearing in the front of VLAN 10 and 20

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Voice	active	Fa0/1
20	Sales	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6

Inter VLAN routing in packet tracer

Each VLAN represents a separate network so to enable routing from one VLAN to another VLAN; we have to either use a layer 3 switch or Router.

On a single router interface, we can create sub interfaces and configure those interfaces to accept traffic from specific VLANs so we can use only one interface that is connected to the Switch for routing purposes. After configuring the sub-interfaces, we have to assign a default gateway on the end device so the default gateway would be the IP address of the sub-interface specified for the same VLAN of the end device.

Now the traffic will travel within Switch when communicating with the end device in the same VLAN and traffic will travel through the Router when communicating with devices on other VLANs.

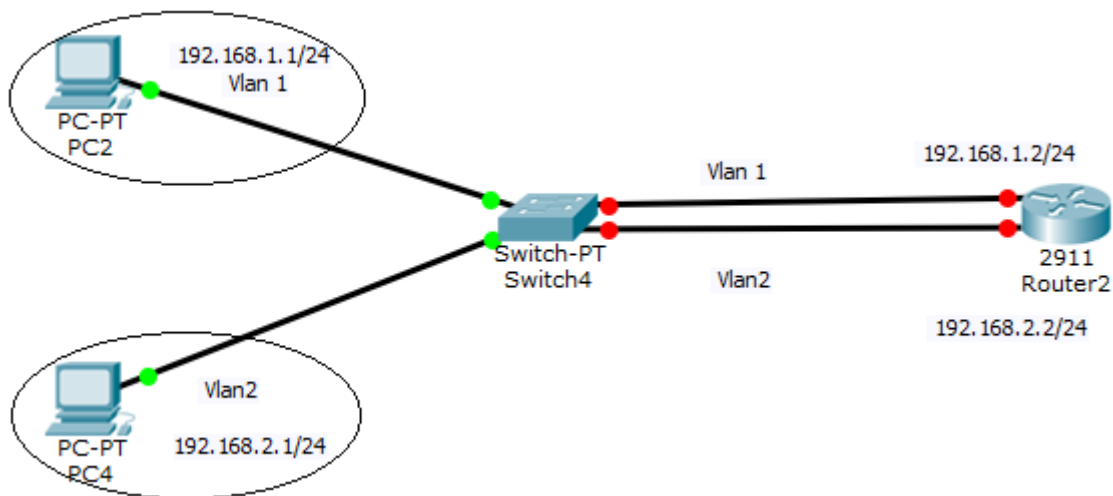
Please check Lab 6.2 router on a stick lab to properly configure inter VLAN routing.

Router on A Stick:

'Router on a Stick' allows routing between VLANs with only one interface. Each VLAN represents a different Subnet. In general, routers can take traffic from only one subnet and transfer it to another subnet. And we can assign only one IP Address to a router interface. 'Router on a stick' allows us to create sub-interfaces, and assign IP Addresses to those sub-interfaces. To make it work, we have to create a trunk connection between the Switch and a router so that traffic from multiple VLANs can be sent to the Router.

If we create a route between VLANs without the 'Router on a Stick' method, then we have to waste interfaces on the switches and routers. And if we enable routing between multiple VLANs then it will become practically inefficient as the switches and the routers will use those multiple interfaces.

The image below is an alternative method for allowing routing between VLANs. As you can see, we are using two interfaces on both the Router and a switch to allow routing between VLANs. We have not created a sub-interface in the below figure.

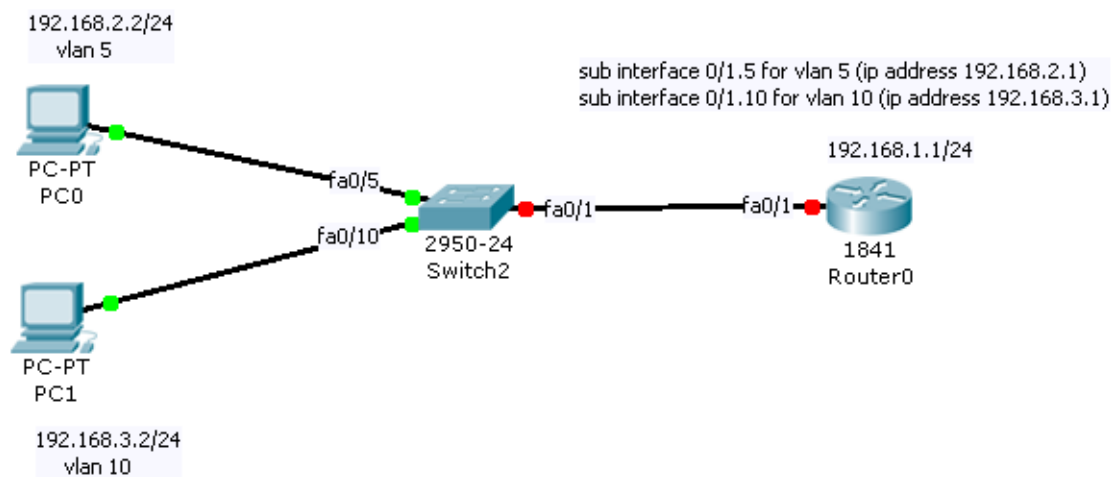


You can see that we have to use extra interfaces for each VLAN. So, it becomes practically non-efficient if we have multiple VLANs. Hence, 'Router on a Stick' is a perfect solution for routing between VLANs with just one router interface.

The simpler way to do routing between VLANs is by using a Layer 3 Switch. We just have to create virtual interfaces for each VLAN and assign them IP Addresses from the same network. A Layer 3 Switch will then enable routing between VLANs as it has routing capabilities as well. However, Layer 3 Switch is quite expensive so it might not be an affordable option for small office networks.

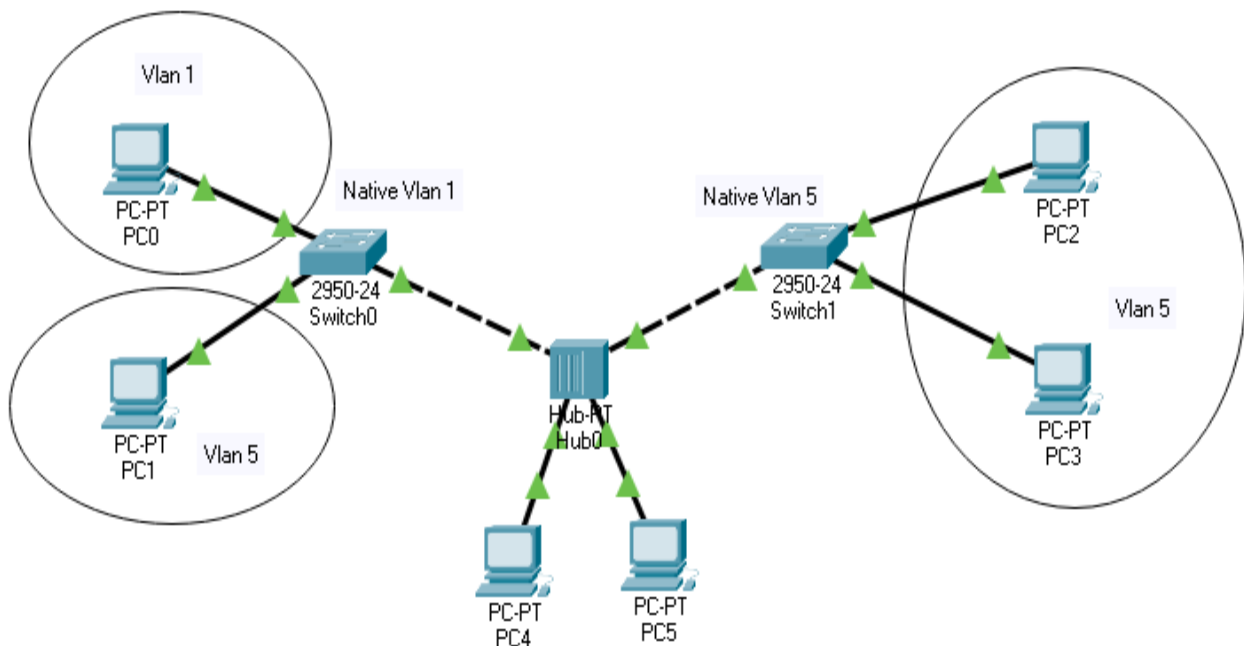
In the below lab, we will configure 'Router on a Stick' that would allow routing between the VLANs. Some of the important concepts in this lab are – to create sub-interfaces, use encapsulation dot1Q command to encapsulate the traffic, and mentioning the VLAN number to ascertain that for which VLAN the sub-interface should respond.

Router on a stick configuration in packet tracer.



Native VLAN and Mismatch:

Native VLAN – Native VLAN allows untagged traffic to be part of one particular VLAN. Native VLAN functionality is used when untagged traffic is received on the trunk connection of the Switch.

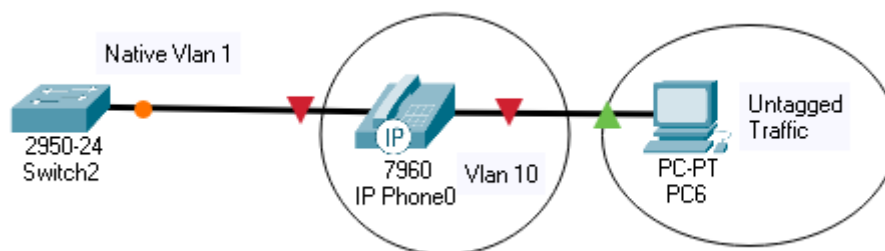


In the above image, you can see that the Hub is connected with the Switch. VLAN and tagging are switching technology and as Hub is unable to tag the traffic then what will happen if PC 4 and PC 5 which are connected to the Hub send the traffic? In that case, we have to specify native VLAN on both of the switches interfaces that are connected to Hub and that will help when the Switch will receive the untagged traffic on those interfaces then it will be the part of the specified VLAN which we have configured as Native VLAN. Although Hubs are not in use now a day's however in a small organization where the budget is low and organizations cannot afford to spend money on new network equipment and they want to use the old network devices then network admin may have to work with such topologies.

In small networks, speed is not the priority for the company so using the device as a hub does not harm the network.

Another network scenario, in which native VLAN is used commonly, is shown below. IP phones are used commonly in the companies these days, they allow daisy-chaining of end devices like PC which saves the interfaces and saves the extra cost of the telecom.

Daisy-chaining the PC is an efficient solution as an organization does not have to buy extra switches to implement IP phones.



PC 6 cannot tag the traffic so we have to configure Native VLAN to route the traffic generated by the PC on the desired VLAN however, Cisco IP phone can tag the traffic so native VLAN will only be used by the PC. By default, native VLAN is configured as 1 on all the interfaces however, in case we want to route traffic to another VLAN then we have to change the native VLAN accordingly.

To check the native VLAN that is configured, we can use the command 'show interfaces trunk'

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    5
```

To configure the native VLAN or to change the default native VLAN, we have to use the following command.

```
Switch(config)#interface FastEthernet 0/24
```

```
Switch(config-if)#switchport trunk native vlan 5
```

We have to enter into interface configuration mode and then use the command shown above; we have to specify the VLAN number where we want to redirect untagged traffic. In the above example, traffic will be routed to the VLAN 5.

Native VLAN mismatch

Native VLAN mismatch happens when interfaces of the Switch are configured with different native VLAN and in that case traffic generated by PC will be routed to both VLAN which will cause issues in the network. The good thing about the Switch is that it detects the native VLAN itself and the following message is displayed on the Command-line interface of the device.

— %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (5), with Switch FastEthernet0/24 (1) —

To fix the Native VLAN mismatch, we have to configure the same native VLAN on all interfaces that are connected to the Hub so the traffic generated will only be redirected to the one VLAN.

In this lab, we have intentionally created a native VLAN mismatch so you can download this lab and can practice fixing the mismatch by assigning the same Native VLAN on both switches.

You will be able to see the mismatch message on CLIs of both of the switches.

Lab Tasks

- Assign IP address and default gateway to the PC
- Create Vlans and assign ports to the Vlans
- Configure trunk connection between Router and a switch
- Create sub-interface and allow routing between the Vlans

Lab Configuration

Task 1

Click on pc0 > click on desktop tab > click on IP configuration > set IP address and default gateway

Task 2

```
Switch(config)#interface fa0/5
Switch(config-if)#switchport access vlan 5
Switch(config-if)#in fa0/10
Switch(config-if)#switchport access vlan 10
```

Task 3

```
Switch(config)#in fastethernet 0/1
Switch(config-if)#switchport mode trunk
```

Task 4

```
Router(config)#in fa 0/1
Router(config-if)#no shutdown
Router(config)#interface fastethernet 0/1.5
Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#in fastethernet 0/1.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 192.168.3.1 255.255.255.0
```

LAB 7: Servers and CDP

Objectives

- Using CDP protocol to find out neighbor information in Cisco packet tracer.
- Setting up a DHCP server in Cisco packet tracer.
- Configure a web server in Cisco packet tracer and access the internet in packet tracer.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

CDP Protocol:

CDP is a handy protocol. It is a fantastic tool for searching the details about directly connected devices. However, being Cisco propriety, it only works if the connected device is a Cisco device. It sends messages to the directly connected devices, these messages contain details about the host, devices, network, IOS, ports, etc. It is enabled by default on all Cisco devices.

CDP protocol generates messages every 60 seconds, these messages are received by the neighbor device and then the information is stored in the database.

CDP provides us with information that is quite useful if we do not have a complete diagram or have incomplete documentation about the network.

By using CDP we can find out the following information:

The hostname of the remote devices

Local and remote interfaces on which the devices are connected

Type of device and the platform

The IP address of the connected devices

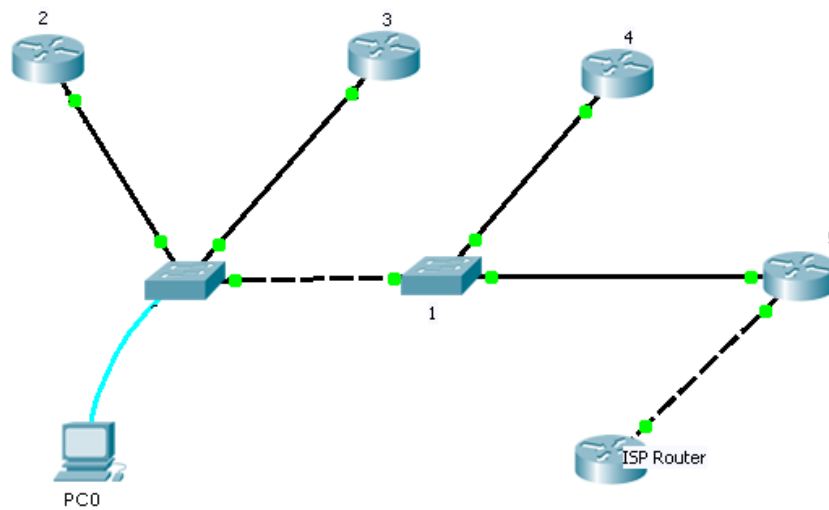
The version of IOS running

Although CDP provides good information about the directly connected devices however it can also pose a security threat to the network as hackers can identify and gather information very easily if any device in the network is compromised so it is the best practice to use this protocol very carefully and disable this protocol on highly secure devices. For e.g, we should not send CDP messages on the interface that is connecting outside of our network.

If we do not have the cisco device on the other end then the CDP protocol will not work however we can use the LLDP (Link Layer Discovery Protocol) as an alternative since it is a non-proprietary protocol so most of the vendors support this protocol. It is an industry-standard protocol with the same functionality.

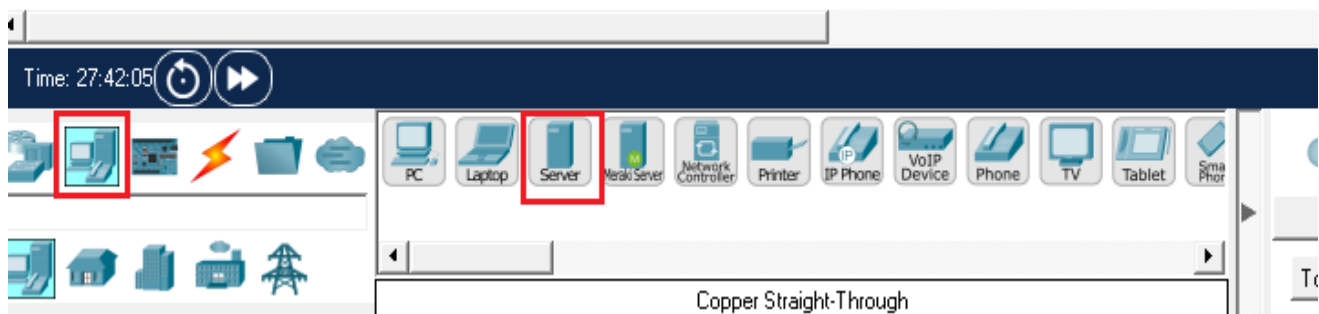
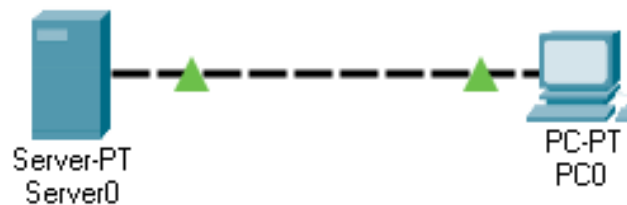
In this lab, you will be required to find information about directly connected devices. To find complete information and details of all the devices, you have to use telnet because CDP can only provide information about directly connected devices. In the below lab, 'device0' does not have a direct link to all Cisco devices. And finally, we have to disable the CDP on the fast Ethernet 0/5 interface of router 5 – which is connected with the ISP router – because we do not want to pass our internal network information to the internet service provider (ISP).

CDP in packet tracer.



DHCP Server:

Setting up a DHCP server and configuring the Router as DHCP server in Cisco packet tracer.



In the Cisco packet tracer, various endpoints are given that we can use to create networks and one of them is the server. This single server provides lots of services like HTTP, FTP, email, DHCP, etc.

Before enabling the DHCP service on the server, we must assign the static IP address to the server to establish connectivity.

The image shows two windows from a network management application. The top window, titled 'IP Configuration', shows the 'FastEthernet0' interface with the 'DHCP' radio button selected. The bottom window, titled 'Server0', shows the 'Services' tab with the 'DHCP' service turned 'On'. It displays configuration fields for the DHCP service and a table of existing IP pools.

IP Configuration Window:

- Interface: FastEthernet0
- IP Configuration:
 - ☒ DHCP
 - ☐ Static
 - IPv4 Address: []
 - Subnet Mask: []
 - Default Gateway: []
 - DNS Server: []

Server0 Window - Services Tab:

DHCP Service Configuration:

- Interface: FastEthernet0
- Service: ☒ On ☐ Off
- Pool Name: serverPool
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0
- Start IP Address: 192.168.2.1
- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 254
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

IP Pools Table:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Pool1	0.0.0.0	0.0.0.0	192.168.2.1	255.255.255.0	10	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.1	255.255.255.0	254	0.0.0.0	0.0.0.0

PC should be configured to accept IPs from the DHCP server so we will change IP configuration from static to DHCP.

To configure the DHCP service, we have to click on the DHCP tab and turn on the DHCP service.

We can change the default name of the pool which is server pool.

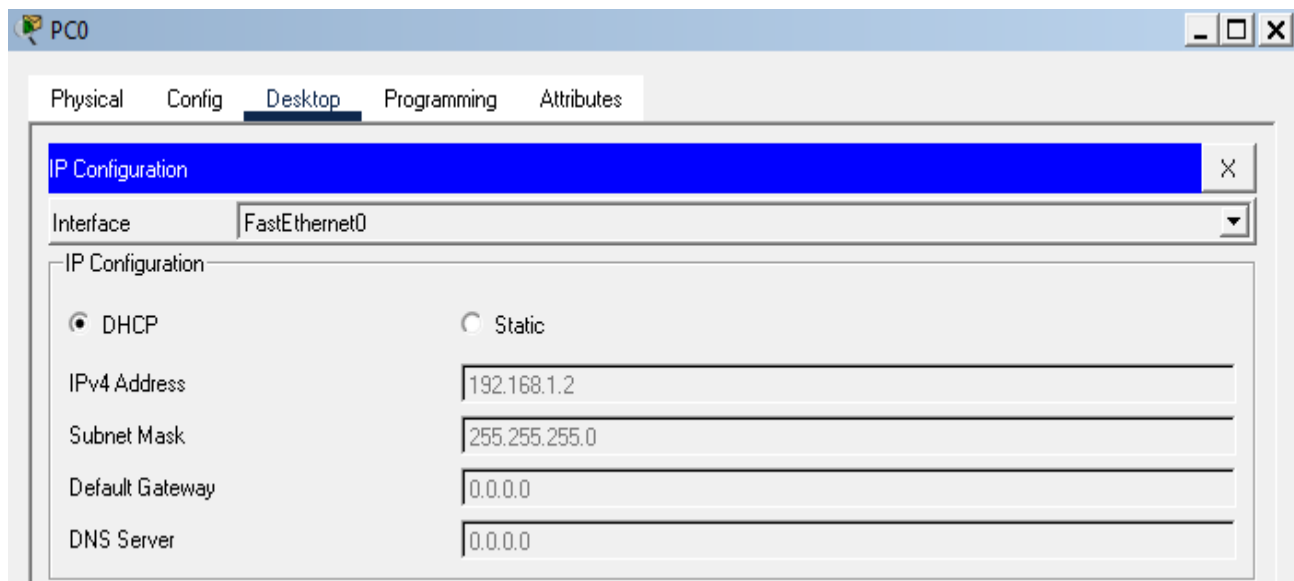
We can set up the default gateway and DNS if required as per our network design.

Define the start IP address so the server will issue the IP address according to our scope mentioned.

We now have to define the maximum number of users so the server will not lease IP addresses to the devices if the limit is reached. In this example, the defined limit is 10 in pool1, now the server will only release 10 IP addresses even if the mentioned IP range scope has a high number of IP addresses.

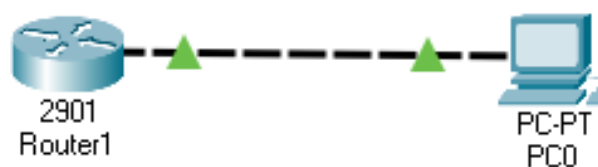
Multiple pools can be added to the server and if any pool is not required then it can be removed easily.

One extra pool name as pool1 has been added to the server with a different IP range so now the server has a lot more IPs to assign.



We can see in the image above that PC has been assigned an IP address successfully by the DHCP server. An assigned IP address is from the configured scope on the server.

Now, as we are done configuring the DHCP on the server. We will configure the Router as a DHCP server in the following packet tracer lab.



We have to configure the same parameters on the Router as well however we have to use the Router's command-line interface to set up the routers with DHCP capabilities.

We will configure DHCP on the Cisco router with 4 easy steps.

Creating a pool and naming the pool. In this lab, we have given the name 'dynamic' to the DHCP pool.

Command : Router(config)#ip dhcp pool Dynamic

Defining the ranges of the addresses that the Router will lease to the clients.

Command: Router(dhcp-config)#network 192.168.1.0 255.255.255.0

Excluding the range of IP addresses that we do not want the Router to lease, this comes in handy when we want to keep some IPs that can be assigned to the devices statically whenever required in our network. In this example, we have assigned IP 192.168.1.1 to the Router so we should exclude this IP to prevent any IP address conflict.

Command: Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5

Finally, we will enable the DHCP service on the Router using the following command.

Command: Router(config)#service dhcp

Additionally, the time limit can be defined for the lease. For e.g, if we want the Router to lease IP for a certain period then we can configure the same using the following command.

Command: Router(config)#lease 10 10 10

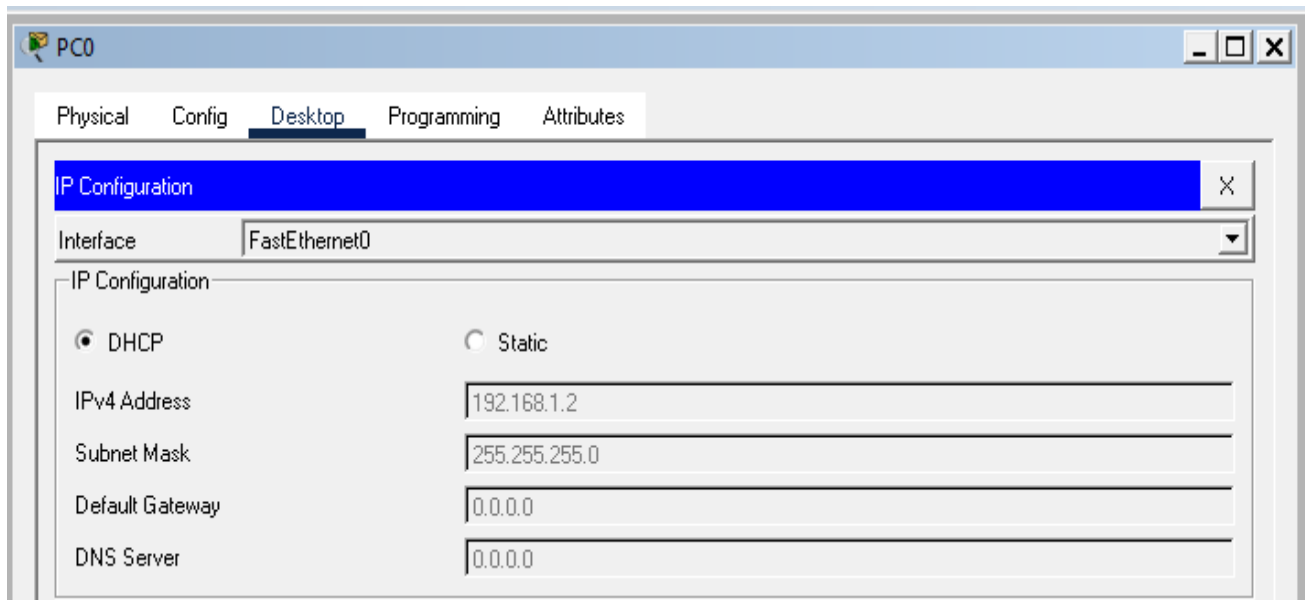
This would mean that Router will lease the IP for 10 days 10 hours and 10 mins. After the defined period, the client will be given a new IP address.

Or

Use the following command to lease the IP for an indefinite time.

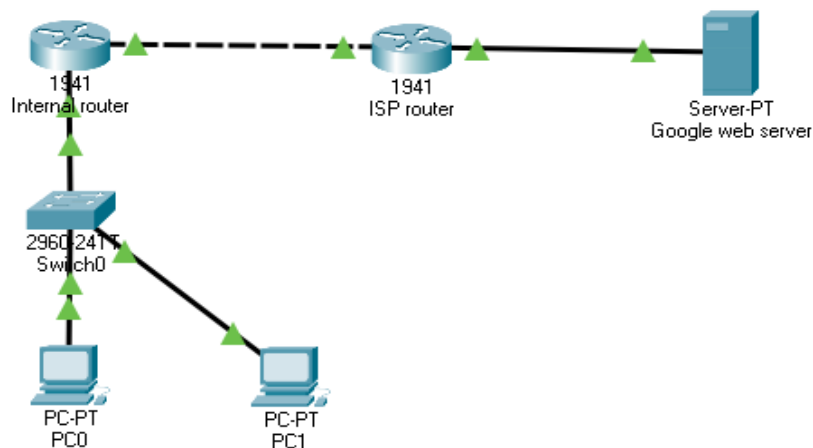
Command: Router(config)#lease infinite.

In the image, we can see that the PC has received the IP address from the Router so now the Router is acting as a DHCP server.



Web Server and Internet:

To simulate the internet, we have to configure the server endpoint given. The server has the capabilities to provide the web service.

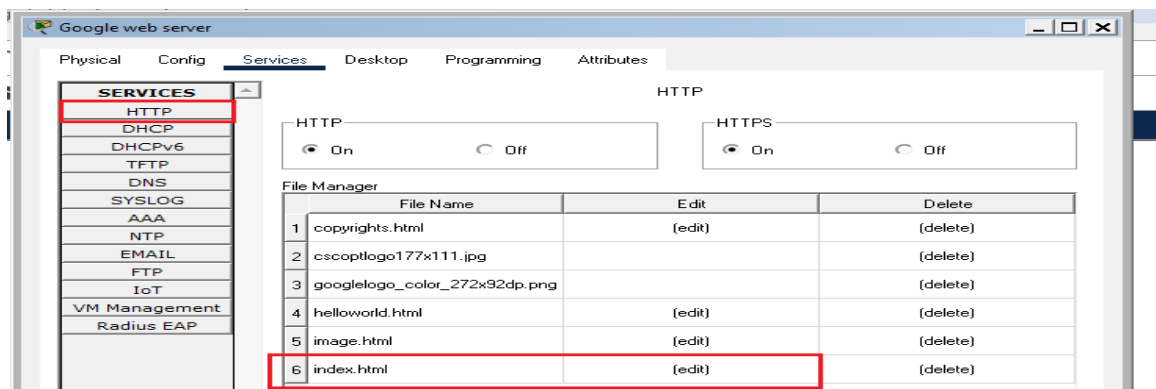


We have two hosts that will be able to access the internet via an ISP router. Our internal network is connected to the ISP router. The interface of the internal Router which is connected to the ISP router has been given public IP address. This is static IP that has been configured on the interface.

ISP router is further connected to the Google web server so after the successful configuration of our network, we should be able to access the google.com

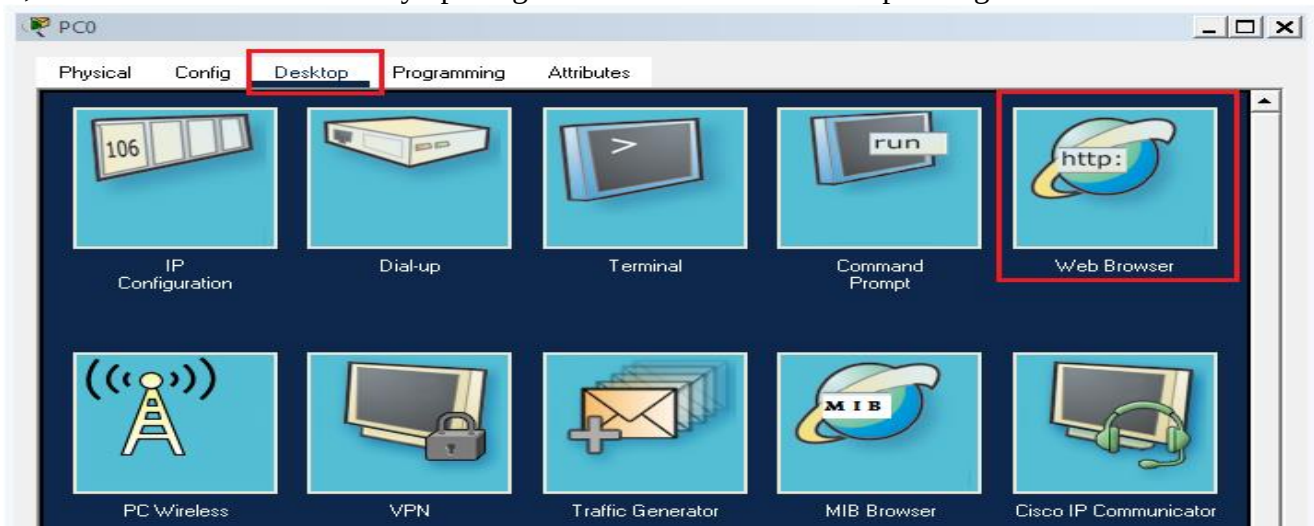
How to configure the web server.

We have to open services and click on HTTP service. This service is enabled by default and we can see that server is hosting some files already. These files are present by default and when we point the web browser to this server, this server will serve the web page to our browser.

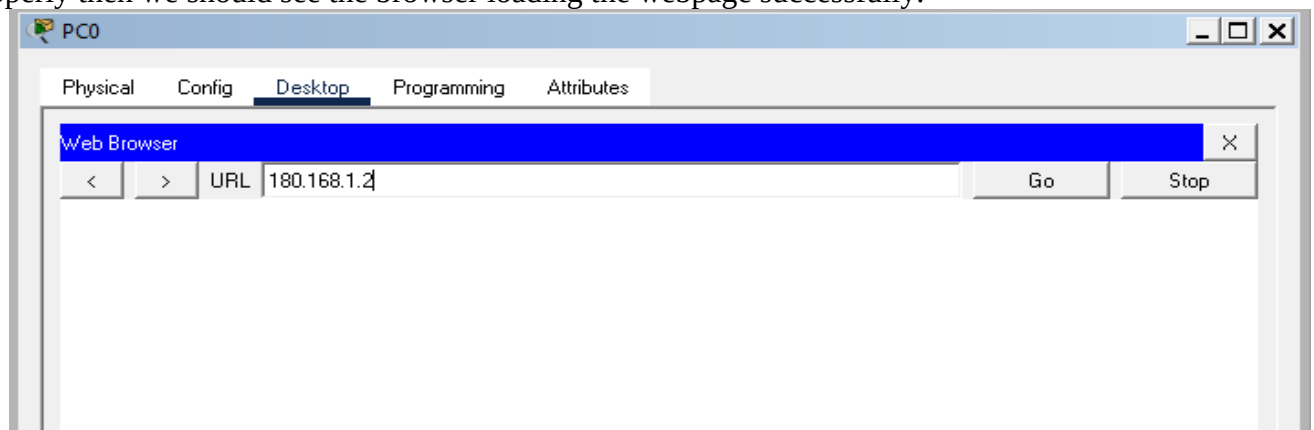


We have configured this server as a Google web server and it is serving a Google web page. To change the content of the webpage, we have to edit the index file present in the web server.

For testing purposes, we can copy the source code of any side that you want the web server to show and paste it into the index file. After saving the file, the web server will show sites according to the source code. Now, we can test the web server by opening the browser on the PC and pointing it to the server.



We have to enter the IP address of the web server in the URL and press enter. If everything is configured properly then we should see the browser loading the webpage successfully.



Before opening the browser, we must make sure that the connectivity of the server is fine.

Download the lab and test the Google web server and if you want the server to host any other site then change the index file present on the server.

This lab is the smaller presentation of the big network like the internet. This is the way how the internet works however there are thousands of routers and servers connected forming a huge network.

You can try adding more servers with different sites and access those sites on the browser of the PC.

DNS in Cisco packet tracer and how to configure DNS on Cisco router

There are a couple of ways in which we can configure dns in packet tracer. We will configure dns on the cisco router and endpoint server.

DNS means domain name server, and DNS contains a database of domain names and IP mapping. DNS servers are very helpful as we don't have to remember the IP address and we can use the domain name instead. Names are much easier to type and learn so DNS plays an important role in every network.

When we enter google.com in the web browser the DNS server finds out the mapped IP of that domain and the HTTP request is forwarded to that IP address.

Configure DNS on the Cisco router

To configure the Cisco router as DNS, we have to enable the DNS service on the Router using the following command.

```
Router(config)#ip dns server
```

Now, we have to map the names with the IP address using the following command.

```
Router(config)#ip host PC1 192.168.1.5
```

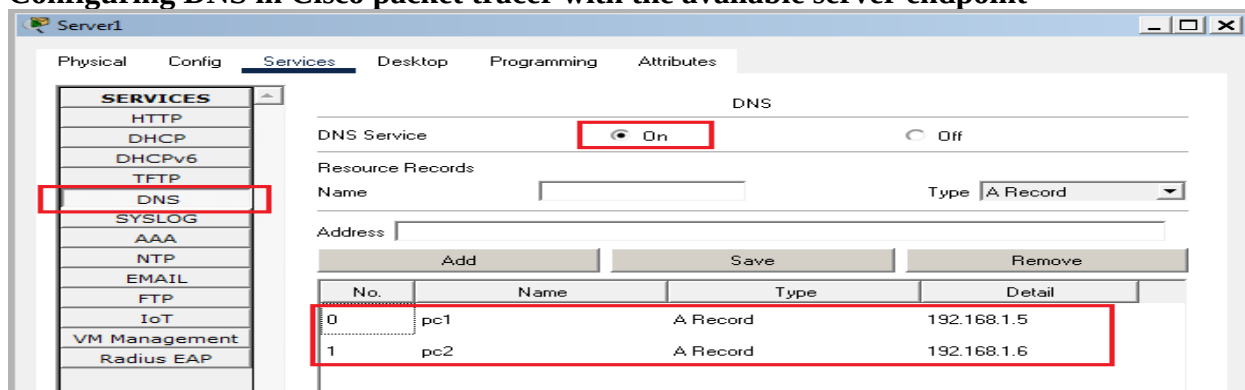
```
Router(config)#ip host PC2 192.168.1.6
```

Finally, we need to configure the DNS server IP in the PC setting. In this case, as the Router is configured as a DNS server, we will use the Router's IP.

Now, we can ping one PC from another PC with their names.

Please note that Configuring the Cisco router as a DNS server is not possible in packet tracer as it does not provide this functionality however If you want to try this lab then you can use GNS3 as it has all the commands.

Configuring DNS in Cisco packet tracer with the available server endpoint



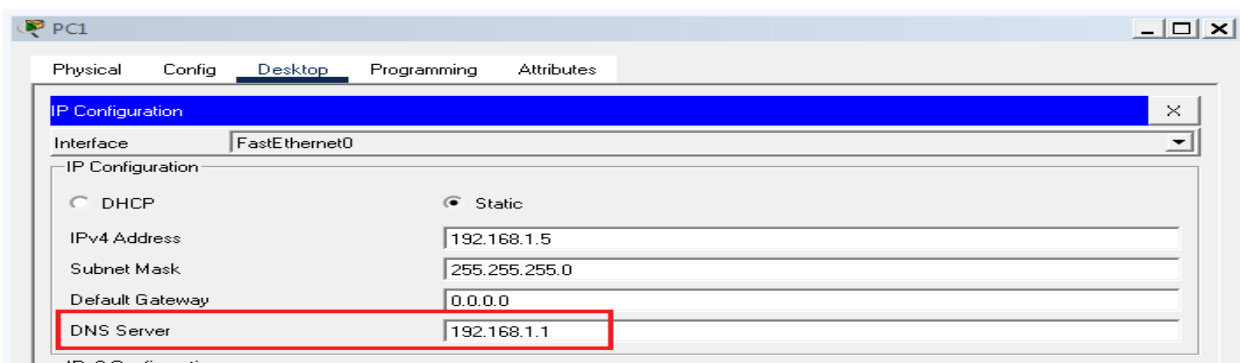
To configure the DNS server, we have to enable the DNS service in the server which is disabled by default.

Now, we have to add an A record or create a name for IP mapping in the server.

Finally, PCs should be configured with the IP address of the server in the DNS settings.

That is all required to successfully enable the DNS server in packet tracer.

Now, we should be able to ping the PCs with the names.



C:\>ping PC2

Pinging 192.168.1.6 with 32 bytes of data:

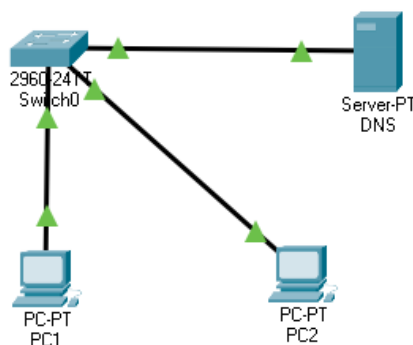
Reply from 192.168.1.6: bytes=32 time=35ms TTL=128

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

As expected, we can ping one PC from another with the names.



Download this lab and ping the PC with names, and you will see the successful ping replies.

Activity for you

Create a mapping for the Switch in the server so that PCs can ping the Switch with the name 'switch'

Lab Tasks

- Use CDP commands to find out Cisco devices 1 to 5
- Find out their IPs and interfaces on which they are connected.
- Disable CDP on interface FastEthernet 0/1 of device 5

(YOU ARE ONLY ALLOWED TO ACCESS SWITCH 0 THROUGH CONSOLE CONNECTION)

(USE TELNET TO ACCESS DEVICE 1)

(TELNET PASSWORD – cisco)

(Privileged mode password – cisco)

Lab Configuration

Task 1

Router#show cdp neighbors

Router#show cdp neighbors detail

Task 2

From Switch 0

local interface – Remote interface and IP address

- | | |
|---------------------|---------------------------------|
| 1. FastEthernet 0/1 | FastEthernet 0/1- 192.168.1.100 |
| 2. FastEthernet 0/2 | FastEthernet 0/0- 192.168.1.3 |
| 3. FastEthernet 0/3 | FastEthernet 0/0- 192.168.1.4 |

From Switch 1

- | | |
|---------------------|-----------------------------------|
| 4. FastEthernet 0/2 | FastEthernet 0/0- 192.168.1.45/24 |
| 5. FastEthernet 0/3 | FastEthernet 0/0- 192.168.1.43/24 |

Task 3

Router(config)#interface fastethernet 0/1

Router(config-if)#no cdp enable

LAB 8: Switch Redundant connections and Loops (STP)

Objectives

- Using CDP protocol to find out neighbor information in Cisco packet tracer.
- Setting up a DHCP server in Cisco packet tracer.
- Configure a web server in Cisco packet tracer and access the internet in packet tracer.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

Spanning tree protocol

In a typical network topology, we have redundant connections between switches. Redundant connections play a very crucial role as it eliminates the single point of failure in the network. However, redundant connections create loop in the network. And to prevent those loops in networks the Spanning Tree Protocol chooses the best link while blocking the redundant links.

Root Bridge is the most important Switch in a Spanning Tree Network. And all the other switches choose the best way to reach a Root Bridge and block the redundant links. Therefore, it is very important to choose the best Switch in the network as a Root Bridge.

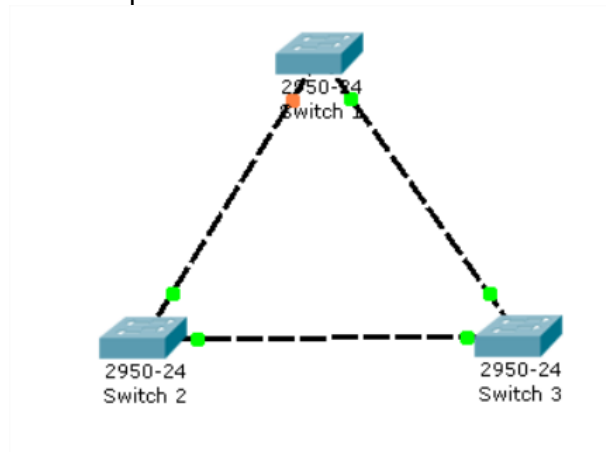
Root is selected on the basis of a Bridge ID. So, whichever Switch will have the lowest Bridge ID, that very Switch will be selected as a Root Bridge. Basically, Bridge ID is made up of a priority number and the MAC address. And by default, all switches have the same priority number – 32768 to be precise – so the Spanning Tree relies on a MAC address for the selection of Root Bridge. But the problem is that by default any switch which has the lowest Bridge ID can be automatically selected as a Root Bridge. And if that Switch is slow then it will slow down the entire network because its network traffic will pass through that Switch. Hence, it is very important that every Spanning Tree Network has the best Switch as a root.

By default, the Spanning Tree is enabled on the switches so if we create a redundant connection on switches then the Spanning Tree Protocol will automatically come into action to prevent a loop in the network. Therefore, for maximum optimization it is very important to select the right Switch as a Root Bridge.

We cannot however, change the MAC address of a switch so we will have to change the priority number of switches to influence the selection of a Root Bridge.

In this lab, we will try changing the priority of a switch to be able to select the Switch of our choice as a Root Bridge. Root Bridge is selected as per the VLAN number so we have to mention that for which VLAN, the Switch is a root. We will also enable a newer version of the Spanning Tree which is a Rapid Spanning Tree.

Spanning tree protocol configuration in packet tracer.



Lab Tasks

1. Select switch 1 as a root bridge
2. Enable rapid spanning tree protocol

Lab Configuration

Task 1

```
Switch(config)#spanning-tree vlan 1 priority 0
```

or

```
Switch(config)#spanning-tree vlan 1 root primary
```

Task 2

All Switches

```
Switch(config)#spanning-tree mode rapid-pvst
```

LAB 9: Setting up basic Router Configurations

Objectives

- Configure the hostname of a router.
- Configure the password for a router.
- We will configure three Cisco routers. We will enable these routers for dynamic routing using the Rip protocol.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

Setting up router name and passwords

By default, Cisco router is configured with the hostname as 'router' and Switch is configured with the default hostname of 'switch' so we must change the default names of all the devices to make it relevant to the network diagram and this also makes it easy to understand and help us recognize the device when we are configuring the device.

It is the best practice to give hostnames to the devices according to the location for example if the Router is situated in network then we can give the hostname as network followed by the model number [Network 1700] or we can give the hostname as per the role of the device for example If a router is connecting to the ISP then a good hostname can be ISP router.

Once a password is configured, users will be prompted to type in that password whenever they enter the Privileged Exec Mode. And if both the 'Enable Password' and 'Secret' are configured, then the users will have to access the Privileged Mode with 'Enable Secret' as enable secret is considered a superior password.

In this lab, we will also configure the Console line password and Auxiliary line password. To set up the Console line and Auxiliary line passwords we have to get under the line configuration mode of both lines.

We will also edit the hostname of our device. The device's name will instantly change when we enter the command – 'hostname *device-name*' on the Command-line Interface. And to check the configured password for the privileged mode you have to exit back to the User Exec mode by simply typing Exit while on the Privileged mode. Then, you will have to type the password every time you enter into the Privileged mode.

You can check the password by initiating a Console connection once again. However, the Auxiliary line password cannot be simulated in the Packet Tracer.

Basic router configuration in packet tracer

Configuration of RIP protocol is very easy as compared to some of the link-state routing protocols like OSPF or EIGRP however RIP protocol does not offer that many features and RIP has a very simple working mechanism. RIP protocol uses hop counts when directing the traffic in the network.

RIP protocol is still widely used due to its simplicity and it is favored in small company networks where there is a fewer number of routers used and small networks do not even need advance routing protocols as

they rarely use features that are provided by the advance routing protocols while big companies with hundreds of Router and complex networks use advance routing protocols as they provide more features that are required for the stability and the functioning of the big networks so at the end it comes up to the requirement of the network.

To set up routers, we will assign an IP address to the router interfaces. It's the best practice to give a unique name to the Router, as well as describing the interfaces of the routers.

To enable the routing between routers we will use a basic routing protocol called Rip. After implementing the Rip protocol on all routers, we can look into the routing table of every Router to check if it has learned the routes with the help of the Rip protocol. We can check the routing table using the command 'show IP route'.

After successful configuration of the routing protocol, we must see the routes in the routing table learned with the help of RIP and if we are unable to see the routes then we have to check the configuration and troubleshoot the issue with the configuration.

In the below image, check out the routes that are learned using RIP protocol by the Router, you should also make sure that Router should learn routes of all the networks in the diagram.

```
Router#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

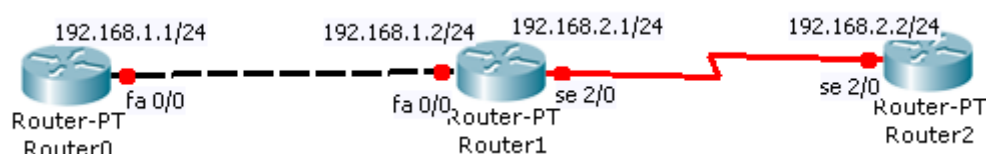
C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
R    192.168.3.0/24 is possibly down, routing via 192.168.2.2, FastEthernet0/1
R    192.168.4.0/24 is possibly down, routing via 192.168.2.2, FastEthernet0/1
```

R is representing the RIP protocol so all the routes having R in the front are learned by the Router via RIP protocol.

By default, the RIP protocol auto summarizes the routes so we have to disable the summarization on each Router.

We will also disable auto summarization of the routes in the routing table by using the command 'no auto summary'.

Also, we will set up a MOTD banner on all the routers.



Lab Tasks

- a. Setting router name to R1
- b. Set privileged mode password to cisco
- c. Set privileged mode secret to cisco
- d. Set console line password to lab
- e. Set auxiliary line password to ciscolab

Lab Configuration:**Task 1**

Router(config)#hostname R1

R1(config)#

Task 2

R1(config)#enable password cisco

Task 3

R1(config)#enable secret cisco

Task 4

R1(config)#line con 0

R1(config-line)#password lab

R1(config-line)#login

Task 5

R1(config)#line aux 0

R1(config-line)#password ciscolab

R1(config-line)#login

Lab Tasks:

- a. assign IP address the router interfaces
- b. enable rip protocol and disable auto summarization
- c. set router 0 name as Arizona, router 1 as Virginia, router 2 as NY
- d. Set description of the interfaces as follow
- e. fa 0/0 interface of router 0 – connection to headquarter

fa 0/0 of router 1 – connection to branch 1

se 2/0 of router 1 – connection to branch 2

se 2/0 of router 2 – connection to headquarter

- f.
- g. Set MOTD on all routers to —- UNAUTHORIZED ACCESS IS NOT ALLOWED —-

Lab Configuration:**Task 6****Router 0**

```
Router(config)#in fa 0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#no sh
```

Router 1

```
Router(config)#in fa 0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no sh
```

```
Router(config-if)#in se 2/0
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#clock rate 56000
```

Router 2

```
Router(config)#in se 2/0
Router(config-if)#ip add 192.168.2.2 255.255.255.0
Router(config-if)# no sh
```

Task 7

Router 0

```
Router(config)#router rip
Router(config-router)#ver 2
Router(config-router)#network 192.168.1.0
Router(config-router)#no auto-summary
```

Router 1

```
Router(config)#router rip
Router(config-router)#ver 2
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#no auto-summary
```

Router 2

```
Router(config)#router rip
Router(config-router)#ver 2
Router(config-router)#network 192.168.2.0
Router(config-router)#no auto-summary
```

```
Router(config)#interface fastEthernet 0/0
```

```
Router(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)#no shutdown
```

Task 8

Router 0

```
Router(config)#hostname Arizona
```

Router 1

Router(config)#hostname Virginia

Router 2

Router(config)#hostname NY

Task 9**Router 0**

Router(config)#in fa 0/0

Router(config-if)#description connection to headquarter

Router 1

Router(config)#in fa 0/0

Router(config-if)#description connection to branch 1

Router(config)#in se 2/0

Router(config-if)#description connection to branch 2

Router 2

Router(config)#in se 2/0

Router(config-if)#description connection to headquarter

Task 10

On all routers

Router(config)#banner motd / —- UNAUTHORIZED ACCESS IS NOT ALLOWED —- /

LAB 10: Setting up Static Routes

Objectives

- Configure the Router for static routing.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

Static Routing Labs

Static routes – Static routes are those routes that are configured manually on the Router. Static routes are useful when we have a very small network with a small number of routers. We do not require routing protocols in a small network where routes are not constantly changing and we do not have new routers added often in the network.

A static route is easy to configure however it takes time to build manual routes so dynamic routing with routing protocol is the only choice we have when working with big networks.

In big networks, we have thousands of routes as they are constantly changing so in that situation, we have to use the advance routing protocol which helps us to easily handle all the routes, and routing protocol also provide us with many features like load balancing, auto alternative route selection, etc

In this example, you will learn to perform the basic configuration of the static route. First of all, we have to assign the IP Addresses to the router interfaces that are connecting different routers. After assigning the IP Addresses, you should easily be able to ping the devices that are directly connected. However, to reach out to the routers that are not directly connected, or to reach to the other networks, routers should have their routes specified in the routing table.

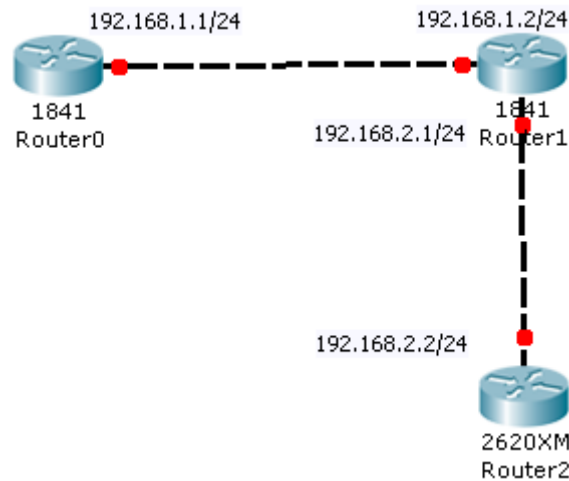
Routers can learn their routes with the help of a routing protocol, or else we can specify the static route on those routers so that they can reach other networks that they don't yet recognize.

And to create a static route, we have to make the Router aware of the other networks so that it can reach out to them. And for that, it has to go out to its interface.

For example, in the following scenario we have to tell Router 0 that to reach the 192.168.1.0 network, it has to go out to its interface using the IP Address 192.168.1.2

Now, we don't have to configure the static route on router 1 as it already knows about the networks 192.168.1.0 and 192.168.2.0 being directly connected to these networks. However, we have to configure the static route on router 2 as it does not know about the 192.168.1.0 network.

Once we have configured the static routes, you will be able to see those static routes in your routing table. And finally, to confirm, you can try pinging router 2 from router 0 and check if they have been configured correctly.



Lab Tasks

- configure ip addresses on all routers and establish connectivity
- create a static route on Router 0 to reach 192.168.2.0/24 network
- create a static route on router 2 to reach 192.168.1.0/24 network
- Test connectivity by pinging router 2 from router 0

Lab Configuration:

Task 1

Router 0

```
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0 Router(config-if)#no shutdown
```

Router 1

```
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.2 255.255.255.0 Router(config-if)#no shutdown
Router(config)#interface fastethernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0 Router(config-if)#no shutdown
```

Router 2

```
Router(config)#interface fastethernet 0/0
Router(config-if)# ip address 192.168.2.2 255.255.255.0 Router(config-if)#no shutdown
```

Task 2

Router 0

```
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
```

Task 3

Router 2

```
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
```

LAB 11: Setting up Default Routes

Objectives

- Configuring default static route in Cisco packet tracer and discuss the role of static route in networking.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

Default Route

A Default Static Route is configured on the routers so the traffic can be transferred to a default route if there is no entry in the routing table for a specific network.

Before sending packets to the other networks, routers check their routing table. The Router drops the packets if it is unable to find a specific route on which traffic should be sent. And when a router does not find the route in its routing table it sends the traffic to a default route, if that default route is configured. So basically, the traffic is sent out of the interface mentioned in the default route in the absence of an entry in the routing table for the traffic.

The default route is most useful when sending traffic to a public network like the internet because it is not possible to have the routing table for the entire internet. Hence, the default route helps the routers to send the packets going out to the internet.

In the local area network, when the destination address is not available in the local network then the data is sent out of the default gateway which is then routed to the other networks by routers; similarly, the default route helps to find the destination which is not available in the routing table.

In the below lab, we will create a default route on the Router to enable routing between different subnets.

The below network scenario is not common in the real world, we have just created this lab to show how the default route can route the traffic without the routing table however we can test the real default route in action if we configure the default route on the Router which is connected to the internet. In the packet tracer lab, we cannot emulate the internet so we are testing the default route in a simple network.

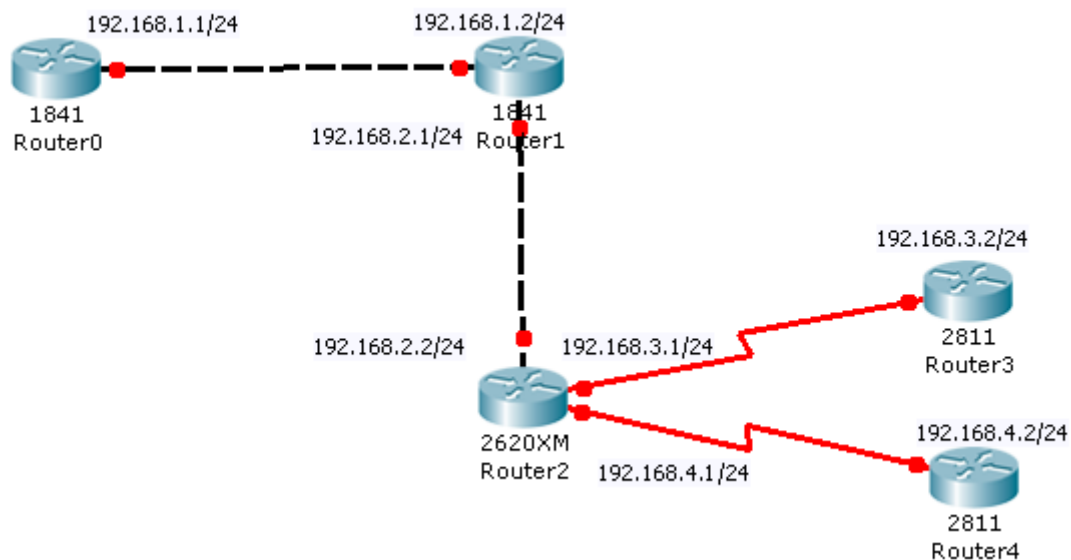
Note: we will only create a default route on the Router. After that, the Router should be able to route traffic. You can ping router 4 from router 0 to check routing in action.

Command explanation – IP route 0.0.0.0 0.0.0.0 192.168.1.2

In the above command 0.0.0.0 means that any IP address with any subnet mask should be sent out of interface 192.168.1.2 if a route is not present in the routing table.

Once we configure the default route, we can see the default route in the routing table with the S* symbol while static routes are shown with just the S symbol.

Before configuring the default route, you will see that gateway of last resort is not set however once the default route is configured, the gateway of last resort is the Router's interface from which traffic will be sent out.



Lab Tasks

- a. Create a default route on router 0 and 1 to reach network 192.168.3.0/24 and 192.168.4.0/24
- b. Create a default route on router 2 to reach network 192.168.1.0/24
- c. Create a default route on router 3 and 4 to reach network 192.168.1.0/24 and 192.168.2.0/24
- d. Test Routing by pinging router 4 from router 0

Lab Configuration:

Task 1

Router 0

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.2
```

Router 1

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.2
```

Task 2

Router 2

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.1
```

Task 3

Router 3

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.3.1
```

Router 4

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.4.1
```

Task 4

Router 0

```
Router#ping 192.168.4.2
```

LAB 12: Configuring WAN Connection

Objectives

- Serial connection explanation and configuration in packet tracer.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

WAN Connection Labs

Serial Connection

The Serial connection is used to create a WAN connection with the help of a service provider who offers a dedicated leased line to its customers.

Business organizations buy leased lines to establish a WAN connection to connect with different offices and business sites that could physically be thousands of miles apart. Businesses have to pay for these leased lines according to the service availed. And accordingly, the service providers charge for this service i.e. based on the bandwidth they offer to their client organizations.

One of the connections used for a leased line is the Serial Connection in Cisco-based networks. If we have Cisco routers on both sides of the network then the default encapsulation known as HDLC will be used by the Cisco routers. However, this will not work with other brands' routers (i.e. with the non-Cisco routers) because this protocol is solely a Cisco proprietary.

If we want to establish a successful connection with a router of other vendors then we must configure the same data link protocol on both routers. If there will be a mismatch then a connection cannot be established and we will see the status of the interface as down under the data link.

If we are configuring a new cisco router then by default the data link protocol would be HDLC however if we are configuring an old device then we should check the configuration and if there is a mismatch then we have to configure the same protocol on both routers.

Although HDLC works out of the box on all cisco routers however it is not widely used because of its proprietary nature and it does not provide much feature so PPP protocol is used by the most organization. Even on cisco routers, organizations prefer PPP protocol over HDLC protocol due to its advanced features and benefits.

In this lab, we will configure a Serial Connection between the routers. We will set up a clock rate at 56000, which will provide a speed of 56 Kbps. In the real world, the DCE side of the cable is configured by the service provider.

Generally, the clock speed is configured according to the availability of the bandwidth speed that the service provider is offering.

The following lab uses a serial cable that will simulate a service provider. We will configure the clock speed on the Router that has the DCE side connected.



Lab Tasks

1. Assign IP address to both routers and turn them on
2. Set clocking on DCE side, which is router 0

Lab Configuration:

Task 1

Router 0

```
Router(config)#interface serial 0/1/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# no shut
```

Router 1

```
Router(config)#interface serial 0/1/0
```

```
Router(config-if)# ip address 192.168.1.2 255.255.255.0
```

```
Router(config-if)# no shut
```

Task 2

Router 0

```
Router(config-if)# clock rate 56000
```

LAB 13: Implementing Router Information Protocol (RIP) ver. 2

Objectives

- Configure the RIP protocol on all the routers.

1. Equipment required: -

- Windows PC
- Packet Tracer

Recommendation

Connectivity with internet

Theoretical Description

RIP Configuration:

Static routing is not feasible in a large network. Hence, to implement routing in an easier way we can use dynamic routing protocols.

Routing information protocol (RIP) is one of the dynamic protocols that can be used for routing. The best part is that Rip protocol is very easy to configure. We will configure Rip version 2 in this lab.

RIP version 1 was introduced in 1988 when networks were comparatively simple that did not require complex routing.

An earlier version of RIP had some limitations hence RIP version 2 was introduced in 1993 which improved some of the features of the RIP protocol.

RIP is a distance-vector routing protocol and hops count to determine the route in the network. Rip protocol only works till 15 hops and it does not work if the network is more than 15 hops away so this prevents RIP to work properly in a big network.

Major differences between version 1 and 2

- RIP V2 supports classless networks, which means we can use VLSM [Variable length subnet mask] in our network while RIP V1 does not support subnetted networks.
- RIP V2 supports authentication while RIP V1 does not support authentication which makes RIP v1 prone to hacking attacks while Rip v2 provides added security while configuring the networks
- RIP V2 uses multicast packets for routing updates while RIP V1 uses broadcast packets which increases unnecessary traffic in the network.

To configure RIP V2 on a router, we just have to type the "version 2" command after the "router rip" command to enable version 2 on any router.

After configuring RIP version 2, we have to advertise the directly connected networks by using the command 'network *network id*' after enabling the RIP protocol on all the routers.

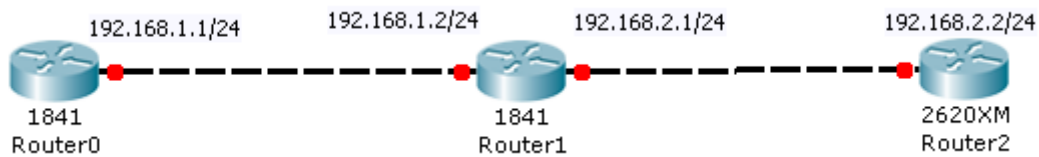
Once the Rip protocol is configured properly you will be able to see the routes in the routing table, learned with the help of Rip protocol by the routers.

Below is an example of a route that is learned after configuring the RIP protocol:

```
R 192.168.2.0/24 [120/1] via 192.168.1.2, 00:00:28, FastEthernet0/0
```

You can verify and check routes in the routing table by using the command 'show IP route'.

We will also disable auto summarization by the Rip protocol.



Lab Tasks

Configure Rip ver 2 protocol on all routers and disable auto summarization.

Lab Configuration:

Task 1

Router 0

```
Router(config)#router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)#network 192.168.1.0
```

```
Router(config-router)#no auto-summary
```

Router 1

```
Router(config)#router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)#network 192.168.1.0
```

```
Router(config-router)#network 192.168.2.0
```

```
Router(config-router)#no auto-summary
```

Router 2

```
Router(config)#router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)#network 192.168.2.0
```

```
Router(config-router)#no auto-summary
```